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Designing for Delegated Digital Health Access: An Interaction Design Study Supporting Informal Caregivers within eHealth Infrastructures

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Abstract

Across Europe, the forty to fifty million people who provide informal care to relatives increasingly do so through digital health systems that were not designed to accommodate them. In most countries, no formal mechanism exists to make that role legitimate. Informal caregivers share credentials, use patients' devices, and attend appointments in person just to access information that should be reachable digitally. This is not an individual failure, nor a technical one. It is a structural condition that existing systems were not designed to see.

This project investigated how interaction with digital health systems can be designed to support the gradual and negotiated delegation of access between patients and their informal caregivers, within eHealth infrastructures where formal proxy mechanisms are absent or inadequate. It was conducted through Research through Design across Spain, Italy, and France, three countries chosen because each represents a distinct stage of the same unresolved problem. Twenty-three participants took part across three phases: contextual inquiry, co-design workshops with patient and caregiver dyads (pairs in which both members of a caregiving relationship participated), and think-aloud evaluation of a medium-fidelity prototype.

Three findings shaped the design. Informal caregivers have built sophisticated parallel documentation systems, handwritten notebooks, shared folders, phone notes, because the official system provides no equivalent. An expert participant who is simultaneously the information architect of the Andalusian Health System and an informal caregiver confirmed that revocation of delegated access is technically trivial and was never designed as a user-facing function not because it could not be, but because no one imagined it would be needed. And most consistently across every complete dyad: the research session was the first occasion on which participants had discussed the delegation of digital health access directly. Across relationships of five, ten, and fifteen years, the conversation had not happened. No system, and no caregiving relationship, had created the conditions for it to occur.

These findings produced a design response covering three interaction flows: configuring delegation, modifying and revoking it, and acting as a delegate while contributing structured observations to the clinical record. Twenty-five design principles were synthesised from the fieldwork, fifteen confirmed across participant profiles, delegation types, and national contexts. The think-aloud evaluation corrected the design in three specific ways: a new screen that resolved a critical perspective-shift failure, the removal of a severity classification that placed clinical judgment on the informal caregiver, and the replacement of pre-selected permission defaults with named starting configurations. In Research through Design, these corrections are not incidental; they are the mechanism through which the prototype produced knowledge that workshops alone could not generate.

The project contributes on two levels. At the level of design knowledge, the twenty-five principles constitute a transferable framework for anyone designing delegated health access. At the level of theoretical contribution, the prototype operationalises Nissenbaum's concept of contextual integrity through mechanisms that existing systems do not provide: a relational identification field that encodes the norm of the

specific caregiving relationship rather than a generic proxy category; a granular permission structure that allows the patient to calibrate access to match the actual norms of that relationship; and an attributed observation channel that preserves the distinction between patient-generated and caregiver-generated clinical data. Two principles document where interface design reaches its boundary: the policy threshold that currently excludes the entire sample from formal delegation rights, and the structural absence of contingency networks that no delegation interface can substitute. The barriers are not technical. They are design decisions that were made without the informal caregiver in the room, and that this project has attempted to make visible and designable.

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1. INTRODUCTION

1.1 Overview of the Problem Domain

Across Europe, digital health infrastructures have expanded rapidly. Citizens can now access medical records, book appointments, and manage prescriptions through national or regional online portals, with authentication handled through national digital identity systems such as BankID in Sweden, SPID in Italy, or Cl@ve in Spain (Muli et al., 2022; Cernadas Ramos et al., 2023). These systems are embedded not only in health services but in banking, public administration, and legal transactions. That level of integration creates both infrastructural efficiency and a structural dependency: the same systems that enable seamless access for most citizens become barriers precisely for those who cannot navigate them alone.

An estimated 40 to 50 million people across Europe provide informal care to elderly or chronically ill relatives, many while in employment (European Parliament, 2024). As health management shifts into digital infrastructures, caregiving increasingly requires digital access. In most cases, however, no formal proxy mechanism exists that would give caregivers legitimate access to the health information of the person they care for. The gap is not peripheral; it sits at the centre of systems that are otherwise highly developed.

The gap is structural, not accidental. Most European patient portals allow parents to manage their children's health records up to a certain age, but no equivalent mechanism exists for adult informal caregivers across most member states. A recent comparative study across Sweden, Norway, Finland, and Estonia identified three types of proxy access in practice: full, controlled, and none, and concluded that further guidance is urgently needed to build a framework that is both ethically sound and GDPR-compliant (Hägglund et al., 2025).

The three countries in which this project conducts its fieldwork each illustrate the gap differently, and together represent three distinct stages of the same unresolved problem. In Spain, the *Ley 41/2002 de Autonomía del Paciente* [Law on Patient Autonomy] permits access to medical records through accredited representation, but only through in-person, paper-based processes; no digital delegation mechanism exists (España, Ley 41/2002). In Italy, a delegation mechanism was introduced by decree in September 2023 as part of the *Fascicolo Sanitario Elettronico* 2.0 [Electronic Health Record] reform, but its regional rollout remains incomplete and uneven (Fascicolo Sanitario Elettronico, 2023). In France, *Mon Espace Santé* [My Health Space] is actively developing a caregiver delegation framework, but no formal mechanism is yet in place (Agence du Numérique en Santé, 2023). The gap is legal in Spain, infrastructural in Italy, and still being defined in France. Table 1 maps this typology across selected European countries.

Table 1. Types of proxy access across selected European countries. Typology (full, controlled, none) follows Hägglund et al. (2025). Country-level data based on Moll et al. (2024), Fascicolo Sanitario Elettronico (2023), España, Ley 41/2002, Agence du Numérique en Santé (2023), and NHS England Digital (2025). Data for Norway and Finland reflects proxy access for minors; formal mechanisms for adult informal caregivers remain unconfirmed. Cell shading indicates implementation maturity: full (light green), controlled (blue), partial or under development (amber), none (light grey).

Country	Proxy access type	Formal mechanism
Estonia	Full	Yes; granular controls with audit trail
United Kingdom	Controlled	Yes; NHS National Proxy Service (mandatory for all GP practices by 2027)
Italy	Controlled (in implementation)	Partial; FSE 2.0 decree (2023), regional rollout ongoing
Norway	Controlled	Partial
Finland	Controlled	Partial
Sweden	None (adult informal caregivers)	No formal mechanism exists
Spain	None (digital)	Paper-based representation only; no digital delegation mechanism (Ley 41/2002)
France	None (under development)	Delegation framework under development; Mon Espace Santé (Agence du Numérique en Santé, 2023)

The authentication infrastructure adds a further layer of complexity. Digital identity systems such as BankID, SPID, and Cl@ve involve multi-step verification processes that many people, particularly older adults, struggle to complete independently (Sveriges Riksbank, 2024; Cernadas Ramos et al., 2023). Research confirms that non-expert users frequently do not understand how these mechanisms work, which creates barriers precisely for those who most need support (Nore & Bernsmed, 2022).

The gap is increasingly misaligned with European policy. Regulation (EU) 2025/327, which established the European Health Data Space and entered into force in March 2025, explicitly grants citizens the right to delegate their health data access to other natural persons (Article 4[2a]), and specifies that digital proxy solutions must align with the European Digital Identity Wallet framework to prevent the proliferation of incompatible national systems (European Parliament and Council of the European Union, 2025). Most member states have neither mechanism in place. The regulation mandates the outcome; it does not prescribe how to design for it.

The gap has consequences at multiple levels: for informal caregivers navigating systems not designed for them, for patients whose autonomy is at stake, for care relationships that evolve over time, and for the design of systems that could respond differently.

The absence of a formal mechanism has not stopped informal caregivers from acting. They call health services on behalf of patients, share login credentials in violation of GDPR, or attend medical appointments in person just to access information they could otherwise receive digitally. This burden is not distributed equally: gender, age, socioeconomic status, and migration background all shape who provides informal care and under what conditions (Togmat Malki et al., 2025). Research on analogous digital contexts confirms this is not an isolated pattern; Latulipe et al. (2022) found that people helping older adults with online banking regularly use the older adult's own credentials, creating serious privacy and security risks. In the health context, the stakes are higher. A comparative study of informal caregiver digital service use in Italy and Sweden found that caregiving intensity and socioeconomic factors significantly shape how, and whether, informal caregivers can access digital support at all (Hassan et al., 2022). For the forty to fifty million informal caregivers across Europe, the workaround is not a failure of the system. It is its de facto operating logic.

The difficulty is ethical as much as technical. Protecting autonomy over one's own health data sits in direct tension with the informal caregiver's need to act effectively on one's behalf. Research confirms that patients are often ambivalent: they may want an informal caregiver to see their medication list but not their mental health notes, and some do not wish to share any of their medical information at all (Latulipe et al., 2018). These tensions do not disappear with a technical solution; they must be designed for. There is also a dimension that bioethics literature tends to understate: shame, loss of control, and the fear of becoming dependent. Any design that does not take those feelings seriously will be resisted before it is ever used.

The problem runs deeper than the legal and ethical layers suggest, because care is not a binary state. Current systems are built around two positions: either the patient manages their own data, or a legal guardian takes over. What they do not account for is the vast middle ground where most actual caregiving happens. The transition into needing help is gradual, personal, and often contested. Research on technology and caregiving shows that care relationships are continuously negotiated as circumstances change, and technologies that do not account for this ongoing negotiation tend to generate friction rather than support (Lazar et al., 2018). Both informal caregivers and care recipients need negotiable agency over how access is configured and used, not just at the point of setup but throughout the caregiving trajectory (Storer & Branham, 2023). The moment when someone starts needing help but has not yet accepted it is the most critical point in the care trajectory. It is also the moment that existing systems are completely unprepared for.

The gap is not a technical limitation but a design and policy choice (Moll et al., 2024). Some countries have already acted on this. The UK's NHS National Proxy Service allows a trusted person to manage prescriptions, appointments, and test results on behalf of a patient, with adoption mandatory for all GP practices by 2027 (NHS England Digital, 2025). Estonia embedded granular access controls into its national eHealth architecture from the start, allowing patients to decide who sees what and logging every access event for transparency (Council of Europe, 2022). These precedents show that the problem is solvable. What is missing is not technical capacity but the interaction design layer that translates policy ambition into lived caregiving practice. This project works in that gap.

1.2 Research Focus

This project originates in three places. The first is intellectual: a previous study on self-care technology for people with Parkinson's disease raised a question that could not be fully pursued within its scope: what happens to digital health access when the person who needs it can no longer exercise it alone, and when the system itself creates the barriers (Barone Cortés, 2025).

The second came through coursework in embodied interaction, grounded in Dourish's (2004) insistence that interaction is always situated in specific bodies and specific contexts. That framing made a quiet question urgent: what does it mean to design digital health systems that assume a body that is always capable, always present, always able to complete the steps? Bodies age. Bodies tremble. Bodies exhaust themselves caring for other bodies.

The third is experiential: the author has personal experience as an informal caregiver navigating a health system that was not designed for that role, an experience that shaped the research focus without determining its conclusions (Núñez-Pacheco & Frid, 2023).

Bringing personal experience into a research process carries methodological responsibilities. When interaction designers work from autobiographical experience, it becomes particularly important to distinguish the designer's voice from the experiences of others (Núñez-Pacheco & Frid, 2023). The fieldwork was conducted as an open listening exercise, not to confirm a pre-existing hypothesis but to understand whether the frictions observed in one context were recognisable across different caregiving situations, relationships, and eHealth infrastructures. Participants were not recruited to validate a position; they were recruited to complicate it. Reflexivity here is not a disclaimer but a methodological commitment that shapes how data was collected, how it is being analysed, and how findings will be interpreted (Braun & Clarke, 2006; Finlay, 2002). This project was shaped by the researcher's own position as a queer person who has spent years navigating institutional systems that were not designed with that body as a reference, and that in some cases actively exclude it. That positionality is not incidental to the research: it is the methodological condition that made it possible to see what CG01, with two decades of experience inside the system, could not see from within it, that the system never generates evidence of the gap because the workaround resolves the operational need without leaving a trace. For the system, nothing is broken. The response here was not empathy but recognition: not designing for people like the participants, but designing with them, from a position that shared enough of the structural experience to understand what the friction cost (Bennett & Rosner, 2019).

The aim of this project is to understand how interaction with digital health systems can be designed to support the gradual and negotiated delegation of access between patients and their informal caregivers, within eHealth infrastructures where formal proxy mechanisms are absent or inadequate. The project pursues this aim through the following research question:

> How can interaction with digital health systems be designed to support the gradual and negotiated delegation of access between patients and their informal caregivers within eHealth infrastructures where formal proxy mechanisms are absent or inadequate?

Three sub-questions guide the inquiry:

SQ1: What workarounds do informal caregivers currently use to access health information on behalf of patients, and where do these workarounds break down?

SQ2: How do patients and informal caregivers negotiate the boundaries and conditions of delegated access over time, and what mechanisms, conflicts, or temporal shifts shape how that negotiation changes as care needs evolve?

SQ3: What interaction qualities, such as granularity, revocability, and transparency, do patients and informal caregivers need from a delegation system in order to trust it and use it on their own terms?

The research question is a design question, not a policy or technical one. Legal frameworks and technical standards can mandate that delegation mechanisms exist; they cannot prescribe what it feels like to use them, or how trust between a patient and an informal caregiver can be expressed, adjusted, and revoked over time. That interaction layer is what this project addresses.

This project contributes to interaction design for health information systems, working at the intersection of three areas: delegated access and consent design, technology for aging and caregiving, and HCI in institutional contexts. Prior work has focused largely on technical feasibility and legal compliance (Hägglund et al., 2025; Moll et al., 2024). This project approaches the problem through design, conducted with informal caregivers and care recipients in Spain, Italy, and France. It contributes to the theoretical conversation around what Nissenbaum (2004) calls contextual integrity: the idea that information should flow in ways that match the norms of the specific relationships it moves through.

The project begins with health information access as its primary lens, while remaining open to the possibility that fieldwork reveals adjacent friction points that expand or reframe the research focus. That openness is not a weakness in the design; it is a condition of doing honest fieldwork (Fallman, 2003).

2. CONTEXT

Designing for delegated health access requires more than a technical solution. It requires a theoretical account of what is at stake: for the patient whose autonomy is in play, for the informal caregiver whose need to act is structurally unsupported, and for the care relationship that evolves between them. This section develops that account across three bodies of work: the ethics of health information access, HCI research on ageing and caregiving, and contextual integrity as a framework for understanding information flows in relational contexts. It then addresses Research through Design as the methodological foundation of the project, authentication infrastructure as a structural barrier to delegation, critical perspectives on designing for whom, and existing design examples that illustrate both the possibilities and the limitations of current approaches to consent and delegated access.

2.1 The Ethics of Health Information Access

Beauchamp and Childress (2019) identify four principles that structure biomedical ethics: autonomy, beneficence, non-maleficence, and justice. In the context of delegated health access, all four are in play simultaneously. Respecting autonomy means not giving informal

caregivers access the patient has not consented to. Beneficence means ensuring that informal caregivers have the information they need to prevent harm. Non-maleficence means recognising that withholding information from an active informal caregiver can itself cause harm, through missed medications, avoidable crises, or decisions made on incomplete information. Justice means that the burden of navigating inaccessible systems should not fall disproportionately on those with fewest resources. What makes delegated access ethically complex is not that these principles conflict in the abstract, but that any concrete design decision about who sees what, under what conditions, and for how long, will inevitably privilege some principles over others. The design task is not to resolve that tension but to make it navigable.

Meslin and Schwartz (2014) show how these tensions play out in electronic health record design specifically. Whether granting an informal caregiver access to a patient's medication history respects or violates autonomy depends entirely on how that access is structured: who can see what, under what conditions, and with what level of patient control. Their analysis makes clear that ethical compliance is not achieved by simply restricting access; it requires designing for granularity, revocability, and transparency. These are not just ethical considerations; they are design requirements.

A dimension that bioethics literature tends to understate is the emotional and relational weight of becoming someone who needs help. Research in HCI and aging consistently documents how older adults resist being positioned as passive recipients of care (Vines et al., 2015). Technologies that frame access as something that happens to a patient, rather than something they actively choose and control, are frequently rejected: not because they fail technically but because they fail relationally. Ethical design in this context is not only about compliance with principles but about preserving the patient's sense of agency throughout the delegation process.

2.2 HCI, Ageing, and the Distributed Nature of Caregiving

Human-Computer Interaction research has increasingly recognised that technology use in caregiving contexts is fundamentally relational and distributed. Lazar et al. (2018) demonstrate that care relationships are not stable states but ongoing negotiations: as circumstances change, so do the boundaries of appropriate access, the distribution of responsibility, and the trust on which delegation depends. Technologies that treat delegation as a one-time configuration rather than a continuous relational process tend to generate friction rather than support. A delegation system that does not account for change over time functions as a snapshot rather than as a delegation system.

Storer and Branham (2023) extend this analysis by showing that informal caregivers and care recipients do not interact with health systems in the same ways, and that both parties need negotiable agency over how access is configured and used. Their work introduces a distinction between supported use, where a patient retains primary control and an informal caregiver assists, and substituted action, where an informal caregiver acts entirely on behalf of a patient who can no longer exercise their own agency. This distinction is not merely descriptive; it functions in this project as an analytical framework. The workshop scenarios in Phase 2 are structured around the full spectrum from supported use to substituted action, and the prototype is evaluated against both ends. A delegation system that works only at one end is not a general solution; it is a partial one. Most current eHealth systems are designed for neither: they assume a single user who is both capable and present. The authentication

architecture of systems such as BankID, SPID, or Cl@ve illustrates this directly: each credential is tied to a single individual, and registering an informal caregiver's device using a patient's credentials can lock the patient out of their own account (Nore & Bernsmed, 2022).

Research on proxy participation in technology contexts raises further design challenges. Latulipe et al. (2018) found that older adult patients are often ambivalent about informal caregiver proxy access: they see its benefits but worry about losing control over sensitive information, particularly data related to stigmatised conditions. This ambivalence is not a problem to be solved; it is a signal that delegation is not a binary decision. Patients do not want to give informal caregivers access to everything or nothing; they want to give access to some things, under some conditions, for some period of time. That is the design space this project works in. If the final design contributes one transferable insight, it is likely to be an operationalisation of this: what it looks like, in interaction terms, to support partial, conditional, and time-bound delegation.

2.3 Contextual Integrity as a Design Framework

The theoretical framework that most directly informs this project is Nissenbaum's (2004) concept of contextual integrity. Nissenbaum argues that privacy is not simply a matter of keeping information secret; it is a matter of ensuring that information flows in ways that match the norms of the specific social context in which it was shared. A patient shares information with their doctor under norms that are very different from the norms under which they share information with their adult child. Violating those norms, even by sharing information that is technically accurate and not secret, constitutes a privacy violation.

Applied to delegated health access, contextual integrity reframes the design problem entirely. The question is not simply whether an informal caregiver should have access to a patient's health records, but what kind of access, to what information, under what conditions, and in ways that respect the specific relational norms of that caregiving relationship. A person helping their parents manage a chronic condition operates under different norms than a professional home carer. An adult child who has power of attorney operates under different norms than a spouse who is a primary informal caregiver. Current eHealth systems are not equipped to reflect these distinctions; they offer binary access or no access, with no mechanism for the kind of contextual calibration that Nissenbaum's framework would require.

This project takes contextual integrity as its primary theoretical lens. The interaction concepts and design principles it aims to produce are not simply interface solutions; they are attempts to operationalise the idea that proxy access should not replicate the patient's full access but should reflect the norms of the specific caregiving relationship: who it is, what it is for, and how it changes over time. That reframing, from access as a technical state to access as a relational and contextual condition, has implications for how patient portals are designed across Europe as the European Health Data Space begins to mandate proxy access for all member states (Moll et al., 2024).

Existing implementations offer partial evidence of what contextual integrity looks like in practice, and where it falls short. Estonia's national eHealth architecture allows patients to define granular access permissions and logs every access event for transparency, which comes closest to the contextual calibration Nissenbaum's framework would require (Council of Europe, 2022). The UK's NHS National Proxy Service takes a different approach: it grants

trusted persons access to prescriptions, appointments, and test results, but without the relational specificity that distinguishes a spouse from a professional carer, or a temporary hospital stay from a permanent arrangement (NHS England Digital, 2025). Both systems solve the technical problem of access. Neither fully solves the design problem of delegation: how a patient expresses contextually appropriate trust, and how that expression can evolve without requiring legal intervention. That gap is what this project aims to address.

2.4 Research through Design as a Mode of Inquiry

Choosing Research through Design as the methodological approach for this project is not a default; it is a decision that the nature of the problem makes necessary. The delegation of digital health access for informal caregivers is a problem that resists resolution through analysis alone. The tensions between privacy and access, between autonomy and dependence, between what systems are designed to do and what caregiving relationships require, cannot be resolved theoretically. They need to be materialised, put in front of people, and explored in context. That is what RtD makes possible (Fallman, 2003; Zimmerman et al., 2007).

In Research through Design (RtD), the designed artefact is part of the contribution. A prototype of a delegated permissions system, a flow that shows how an informal caregiver negotiates access to clinical data, an interface that handles dynamic consent: these are not just outputs. They are the means through which knowledge about how to design in this space is generated. As Zimmerman et al. (2007) argue, RtD is particularly valuable when a problem has no established solution and when the act of designing is the most honest way to explore what a solution might look like.

This also means accepting that the process is not linear. Krogh et al. (2015) identify five methods of experimentation in Research through Design, each describing a distinct way of drifting productively through a non-linear process: accumulative, comparative, serial, expansive, and probing. Rather than treating deviation from a planned path as a failure, these methods frame iterative reorientation as a legitimate and generative mode of inquiry. The fieldwork, the prototyping, and the evaluation are not three sequential stages; they are three ongoing conversations with the same problem. Insights from participants reshape the prototypes; the prototypes surface questions the fieldwork did not anticipate; the evaluation reveals tensions the theory had not named. That iterative structure is not a limitation of the method. It is the method. What it demands in return is rigour in documentation: every design decision, every iteration, every moment of resistance from participants is data. The contribution of this project is not a finished design. It is knowledge about how to design the delegation of digital health access given these specific tensions, in these specific contexts, for these specific people.

2.5 Authentication Infrastructure as a Barrier to Delegation

Digital identity frameworks such as BankID in Sweden, SPID in Italy, and Cl@ve in Spain are designed around the individual: one identity, one credential, one device. This architecture makes delegation structurally difficult. The introduction of eIDAS 2.0 and the European Health Data Space Regulation represents an institutional shift toward cross-border interoperability and delegated access, explicitly requiring proxy mechanisms aligned with the European Digital Identity Wallet framework (European Parliament and Council of the European Union, 2025). What this policy opening does not resolve is the onboarding

experience: the moment at which a patient must actively configure who can access what, under what conditions, and for how long. Research on digital onboarding in healthcare consistently shows that this moment is where trust is either established or lost, and that systems which treat onboarding as a technical formality rather than a relational act tend to generate abandonment rather than adoption (Latulipe et al., 2018). Designing that moment well is not a secondary concern; it is where the delegation system either becomes real for the people who need it or remains a policy aspiration.

2.6 Critical Perspectives: Designing for Whom

Every design decision about who a system is built for, what kind of body it assumes, and what kind of user it imagines carries values, and those values can include or exclude. This project takes that seriously: not as a rhetorical gesture but as a methodological commitment that shapes how the fieldwork was conducted, how participants were recruited, and how the findings will be interpreted.

Dourish (2004) argues that interaction is always embodied and situated in specific bodies, in specific contexts, under specific conditions. This matters for a project about delegated health access because the body of the informal caregiver is rarely the body that eHealth systems imagine. These systems are designed for someone who is young, healthy, digitally fluent, and acting on their own behalf. The informal caregiver who is older, who has limited digital literacy, who is managing their own health alongside someone else's, or who is doing all of this in a second language, encounters a system that was not built with them in mind. Dourish's insistence that computation must be understood in terms of the social and material contexts in which it is used, not as an abstract technical layer imposed on top of them, is a foundational commitment of this project's design approach. This embodied perspective extends beyond the informal caregiver to the care recipient. A person with tremors cannot reliably complete a multi-step authentication process. A person with cognitive fatigue cannot navigate a portal designed for someone operating at full capacity. When the body itself becomes a barrier to digital access, the need for delegation is not a preference; it is a necessity. Designing for delegation is inseparable from designing for bodies that do not fit the system's assumption.

Bardzell (2010) extends this critique by arguing that HCI has historically centred on a particular kind of user: male, able-bodied, technologically confident, and operating in conditions of relative stability. Feminist HCI calls for design that attends to pluralism, participation, and the redistribution of agency: asking not just whether a system works, but for whom it works, and who it leaves out. In the context of informal caregiving, this question is not abstract. The majority of informal caregivers in Europe are women, and the structural conditions under which they provide care, including reduced working hours, interrupted careers, and limited institutional support, are themselves the product of systems that have consistently undervalued care work (Togmat Malki et al., 2025). A design project that does not name these conditions risks reproducing them.These structural conditions shape not only who provides care but whose body the system was designed to accommodate.

Intersectionality, as developed by Crenshaw (1989) and elaborated by Collins and Bilge (2020), offers a framework for understanding how multiple axes of inequality, including gender, age, class, migration status, and disability, interact and compound rather than operate independently. The informal caregivers this project works with are not a homogeneous group: some navigate health systems across national borders and in languages

that are not their own; some are young and digitally capable but institutionally invisible; some are older and excluded by design choices that assume a different user. A delegation system that works for one kind of informal caregiver but not another is not a solution; it is a redistribution of exclusion.

A further dimension that the intersectionality framework requires naming explicitly is the structural invisibility of caregiving relationships that fall outside the heteronormative family model that eHealth systems were designed for. Weston's (1991) concept of chosen family describes kinship networks built through voluntary bonds of mutual support rather than biological descent or legal recognition, a form of family organisation with particular significance in LGBTQ+ communities, where families of origin have not always constituted a safe or available source of care. Research on LGBTQ+ older adults and informal caregiving consistently documents that care in these communities is most often provided by partners, spouses, and friends rather than biological relatives, and that the absence of formal institutional recognition of these relationships creates specific barriers to access and specific vulnerabilities for both informal caregivers and care recipients (Fredriksen-Goldsen et al., 2015; Muraco & Fredriksen-Goldsen, 2011). When a caregiving relationship is not legible to the system, access that functions in routine contexts can fail precisely in the moments of crisis when it matters most. A delegation system that bases formal recognition on legal kinship categories does not solve this problem; it reproduces it. The European Health Data Space Regulation (EU) 2025/327, which grants citizens the right to delegate health data access to other natural persons without specifying the nature of the relationship (Article 4[2a]), opens the design space this project argues is necessary: one in which the criterion for delegation is the patient's active consent, not the system's prior recognition of the relationship through which care is actually given.

These perspectives are not added as a layer of critique on top of the project; they are built into it. The decision to recruit across three national contexts, to include informal caregivers whose relationships fall outside legal kinship categories, to seek out participants navigating health systems across national borders and in languages that are not their own, to include care recipients as active participants rather than passive subjects of the design, and to conduct all interviews in participants' own language: all of these reflect an understanding that the design of delegation is also a design of who gets to participate in their own healthcare. That is a political question as much as a technical one, and this project approaches it as such .

2.7 Design Examples: What Delegation and Consent Look Like in Practice

Theory defines the problem. Design examples show what attempting to solve it looks like, and where those attempts fall short. Three projects are particularly instructive for this work: not because they solve the delegation problem but because each illuminates a different dimension of it.

The concept of dynamic consent, introduced by Kaye et al. (2014), was one of the first attempts to move beyond static, paper-based consent models in health research. At its core, dynamic consent proposes a personalised digital interface that connects researchers and participants continuously, allowing individuals to adjust their consent preferences over time rather than making a single irreversible decision at enrolment. The parallels with delegated health access are immediate: both involve ongoing, revisable relationships between a person

and a system that holds sensitive information about them. What dynamic consent does not address is the relational dimension: in caregiving contexts, consent is not just a matter of individual preference but of negotiated trust between two people whose needs and circumstances are both changing. Dynamic consent gives the individual more control. It does not give the dyad a shared space in which to exercise that control together.

Tseng et al. (2024) extend this further by applying human-centred design to value-centred consent interfaces, demonstrating that standard consent interfaces routinely misalign with patient values because they were designed without patients. As Bardzell (2010) would argue, designing for participation requires first asking whose values the system currently encodes. The limitation is that their work focuses on individual data donation; the informal caregiver is absent, and the relational negotiation that defines informal care has no place in their model.

A distinct form of evidence comes not from a design intervention but from the internal documentation of one of the health systems studied in this project. Access to the official user manuals of the Andalusian Health System (SAS), including ClicSalud v2.11, ClicSalud Asistencial v1.3, the Salud Responde BackOffice system, and the *Sistema de Seguimiento Proactivo de Pacientes Crónicos* [Proactive Monitoring System for Chronic Patients] (SCSP, updated November 2025), reveals a consistent architectural logic across four decades of system development. The most recent of these documents specifies a system that contemplates twelve professional roles, remote patient monitoring, personalised care plans, and bidirectional messaging between clinicians and patients, and contains no role, no field, and no interface element for the informal caregiver. [Figure 1,Salud Responde main menu, showing seven functional categories with no caregiver or delegated user role] illustrates this directly: the main menu of Salud Responde presents seven functional categories, none of which contemplates a delegated user or a caregiver role.

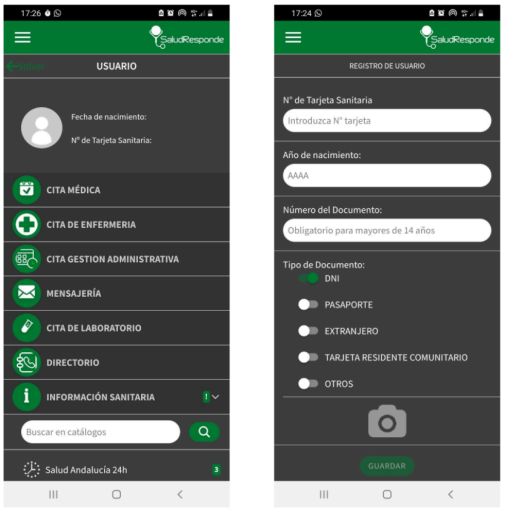


Figure 1. Salud Responde, main menu (left) and user registration screen (right). Personal data has been anonymised.
Source: screenshot provided by CG01, May 2026.

This absence is not incidental. The ClicSalud v2.11 manual states explicitly that clinical information within the system originates exclusively from internal sources and cannot be introduced from external ones (Subdirección de Tecnologías de la Información, 2021), a design decision that structurally forecloses the channel for informal caregiver-generated clinical knowledge that the fieldwork consistently identified as absent. The BackOffice documentation further reveals that when the system cannot resolve a request digitally, the default resolution is physical presence at the health centre counter: a design that collapses toward the in-person precisely at the moments of greatest informal caregiver load. The registration screen of Salud Responde confirms this at the point of onboarding: the only fields are the patient's health card number, date of birth, and identity document [Figure 1 Salud Responde registration screen, showing fields for health card number, date of birth, and identity document only].

There is no field for a representative, a proxy, or a caregiver. What these documents make visible is not a technical limitation awaiting resolution but a sustained design decision: the adult informal caregiver was not imagined as a user of the system; the only form of proxy access these systems contemplate is parental representation of minors, and that boundary is documented in their own specifications. A co-design session conducted with CG01, a participant with over two decades of experience in the information architecture of the SAS, confirms that this invisibility is self-reinforcing in two distinct ways. First, because informal caregivers access the system using the patient's credentials, the system registers their actions as the patient's own and therefore never generates the evidence of a gap that might prompt a design response. Second, the absence of revocation as a user-facing function is not a technical constraint: removing a delegated relationship from the database would be architecturally straightforward. It was simply never designed. The system does not perceive informal caregiving as a problem because the workaround, using the patient's device and credentials, resolves the operational need without leaving a trace. For the system, nothing is broken.

The French system offers a partial counterexample that reveals the same limit from a different angle. Mon Espace Santé allows a user to add profiles for dependants, but the only legal basis it recognises is parental representation of a minor: the onboarding screen requires the user to certify that they are the legal representative of the profile being added [Figure 2 Mon Espace Santé onboarding screen, requiring legal representation certification for minor profiles only]. The Doctolib platform goes further, allowing multi-profile access within a family account, but treats revocation as deletion: removing a profile from the list triggers a warning that all health data associated with that profile will be permanently deleted within thirty days [Figure 3 Doctolib profile management screen, showing deletion warning triggered by profile removal]. Access and data are coupled in a way that makes revocation punitive rather than safe. These screens were documented by CG08 during a live configuration attempt in the contextual inquiry phase, in which four design failures were identified in real time. It is that invisibility that this project attempts to make designable.

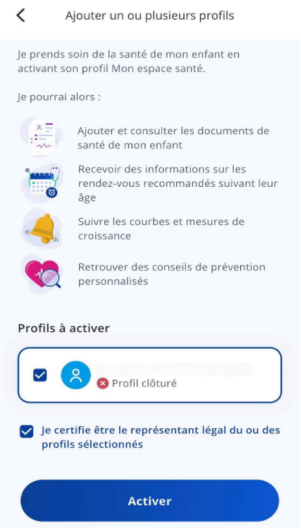


Figure 2. Mon Espace Santé, profile onboarding screen. Personal data has been anonymised. Source: screenshot documented by CG08 during Phase 1 fieldwork, May 2026.

The most directly relevant example is the work of Scerpella et al. (2025) on shared access to patient portals, conducted through the Coalition for Care Partners in the United States. Their study identifies a problem recognisable from the fieldwork: shared access functionality exists in many patient portals, but adoption is consistently low because the interfaces are complicated and most patients and informal caregivers are unaware the feature exists. Their intervention, a step-by-step video tutorial prototype developed through human-centred design, received positive feedback and improved awareness. It is a meaningful contribution. But it also reveals the ceiling of what a tutorial can do: it teaches people how to use a system that was not designed with them in mind. The underlying interface remains designed for a single user. The relational logic of caregiving, the gradual, negotiated, and

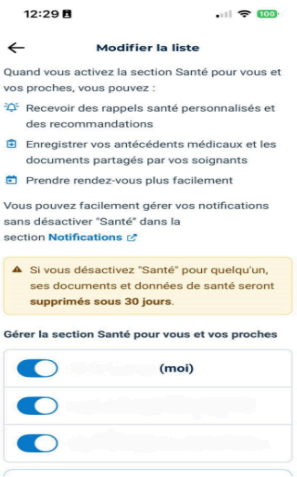


Figure 3. Doctolib, profile management screen showing data deletion warning triggered by profile removal. Personal data has been anonymised. Source: screenshot documented by CG08 during Phase 1 fieldwork, May 2026.

contextually specific nature of delegation, is not something a tutorial can introduce into a system that was not built to support it.

Taken together, these examples trace a trajectory: from consent as a one-time decision, to consent as an ongoing individual preference, to consent as a shared access feature that informal caregivers struggle to find and use. Each step is an improvement. None of them reaches the design space this project works in, where delegation is relational, contextual, and gradual, and needs to be designed for from the start rather than added as a functionality after the act. That gap is not a criticism of these projects. It is the justification for this one.

3. METHOD

The tensions between privacy and access, between autonomy and dependence, between what systems are built to do and what caregiving relationships actually require, cannot be resolved theoretically. They need to be encountered in practice, put in front of people, and explored through design. That is the logic behind adopting a Research through Design approach (Fallman, 2003; Zimmerman et al., 2007), and it shapes every methodological decision that follows.

The methodological toolbox draws on three established approaches in HCI and participatory design research. Contextual inquiry (Beyer & Holtzblatt, 1998) studies people in their own environment rather than in controlled settings, producing the kind of situated knowledge that larger but shallower studies cannot. Experience prototyping (Buchenau & Suri, 2000) allows both researchers and participants to gain first-hand understanding of what a design might feel like in practice; in this project it forms the foundation of the co-design workshops in Phase 2, where participants do not evaluate finished concepts but generate them. Reflexive thematic analysis (Braun & Clarke, 2006) provides a rigorous but flexible framework for identifying patterns of meaning across qualitative data, including affective reactions, hesitations, and moments of resistance that surface-level metrics would miss. These three methods were chosen because they are appropriate for a problem that is relational, contextual, and emotionally complex, and because together they allow the research to move from understanding current practice to designing for future possibility.

The process shares structural similarities with the Double Diamond model: a divergent phase of understanding followed by a convergent phase of designing and evaluating. However, the Double Diamond assumes a degree of iterative access to participants that is not realistic in this context. Informal caregivers operate under conditions of sustained cognitive and emotional load; asking them to participate in multiple rounds of testing and refinement would be both logistically impractical and ethically questionable. The three phases are therefore understood as iterative and overlapping at the level of the research process, but structured and bounded at the level of participant engagement. What the Double Diamond captures in structure, RtD captures in epistemology: the knowledge comes from the doing, not from completing the phases in sequence (Fallman, 2003; Zimmerman et al., 2007).
The fieldwork informs the prototyping; the prototyping surfaces questions the fieldwork did not anticipate; the evaluation reshapes both. The three phases that follow reflect that logic. Figure 4 provides an overview.

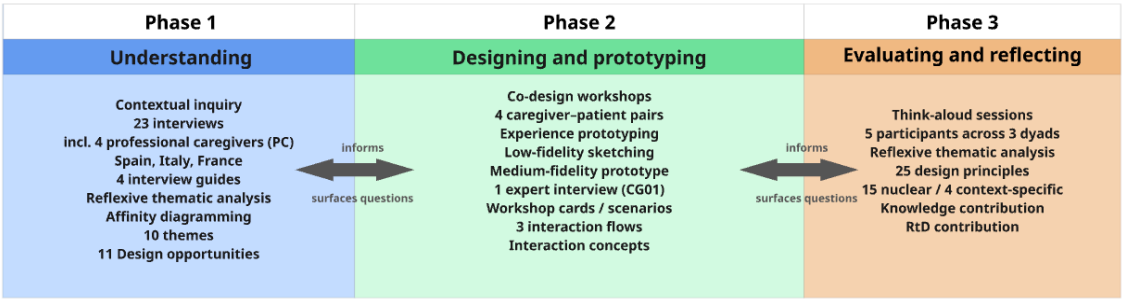


Figure 4. Phases are iterative and overlapping at the level of the research process, but structured and bounded at the level of participant engagement. Phase 2 is represented at greater width to reflect its methodological centrality as the generative core of the Research through Design process (Fallman, 2003; Zimmerman et al., 2007).

3.1 Participants and Recruitment

The study includes informal caregivers, care recipients, and a small number of participants who occupy a hybrid role: professional caregivers who simultaneously provide informal care to family members. This hybrid category is analytically significant because it surfaces a form of expertise that is neither purely family nor purely institutional, and that existing eHealth systems are equally unprepared for. Table 2 presents the full participant sample.
Spain, Italy, and France were selected not as a comparative sample but as three contexts that illustrate distinct stages of the same unresolved problem: a gap that is legal in Spain, infrastructural in Italy, and still being defined in France. Together they offer variation in the nature and degree of the inadequacy that the research question addresses, rather than a geographically representative cross-section of European eHealth systems. Participant recruitment reflected both this conceptual logic and the researcher's existing networks in each country.

Table 2. Full participant sample. Codes follow three roles: CG (informal caregiver), CR (care recipient), and PC (professional caregiver in a hybrid role). Delegation type follows Storer and Branham (2023): supported use indicates that the care recipient retains primary agency; substituted action indicates that the caregiver acts on behalf of a care recipient who can no longer exercise their own agency; hybrid indicates a contested or intermediate position. *Dyad completeness* indicates whether both members of a caregiving relationship were independently interviewed. Shaded rows identify complete dyads. Several care recipients were unable to participate directly due to severe cognitive or physical conditions; in these cases, caregivers acted as proxy informants, as described in Section 3.3. Cell shading indicates whether both dyad members were interviewed: yes (light green), no (light grey).

Code	Age	Role	Country / Health system	Delegation type	Both dyad members interviewed
CG01	44	Informal caregiver	Spain / SAS	Supported use	Yes
CG02	45	Informal caregiver	Spain / SAS	Supported use	No
CG03	39	Informal caregiver	Spain / Asisa (private)	Supported use	No
CG04	28	Informal caregiver	Spain / SAS	Substituted action	Yes
CG05	38	Informal	Spain / SAS	Substituted action	Yes

		caregiver			
CG06	35	Informal caregiver	Spain / SAS + Quirón (private)	Substituted action	No
CG07	74	Informal caregiver	Spain / SAS + Asisa (private)	Substituted action	Yes
CG08	43	Informal caregiver	France / Assurance Maladie	Hybrid	Yes
CG09	43	Informal caregiver	Italy / SSN	Supported use	No
CG10	56	Informal caregiver	Italy / SSN	Substituted action	No
CG11	33	Informal caregiver	Italy / SSN ASL	Substituted action	Yes
CR01	73	Care recipient	Spain / SAS + Asisa (private)	Substituted action	Yes
CR02	91	Care recipient	Spain / SAS	Substituted action	Yes
CR03	29	Care recipient	Spain / SAS	Substituted action	Yes
CR04	53	Care recipient	Spain / SAS	Substituted action	Yes
CR05	64	Care recipient	Italy / SSN ASL	Supported use	Yes
CR06	76	Care recipient	Italy / SSN	Substituted action	Yes
CR07	41	Care recipient	France / Assurance Maladie	Hybrid	Yes
CR08	84	Care recipient	Italy / SSN Roma	Supported use	No
PC01	54	Professional caregiver / hybrid	Spain / SAS	Substituted action	No
PC02	36	Professional caregiver / hybrid	Spain / SAS	Supported use	No
PC03	57	Professional caregiver / hybrid	Italy / SSN	Substituted action	Yes
PC04	53	Professional caregiver / hybrid	Italy / SSN CPU Piamonte	Substituted action	No

Delegation type is classified following the spectrum developed by Storer and Branham (2023) and illustrated in Figure 5: a dynamic continuum whose positions are not fixed but may shift over time as health conditions and care needs evolve.

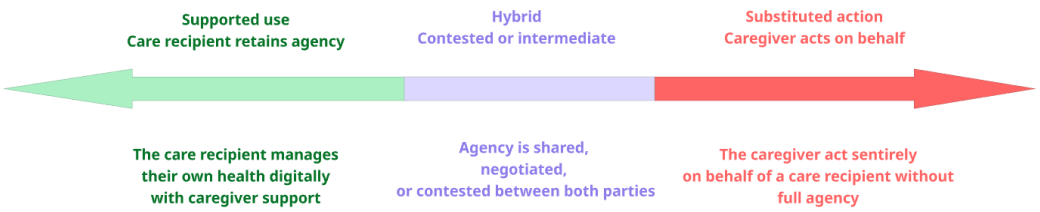


Figure 5. The delegation spectrum as an analytical framework. Adapted from Storer and Branham (2023). The spectrum is not a fixed typology but a dynamic continuum: a caregiving relationship may shift position over time as health conditions and circumstances evolve.

In Spain, the primary digital health platform accessed by participants was ClicSalud, the citizen-facing portal of the Andalusian Health System (SAS), whose internal architecture is analysed as documentary evidence in section 2.7. In Italy, participants navigated the *Fascicolo Sanitario Elettronico* [Electronic Health Record] (FSE) and its regional implementations, including the SSN portal in Rome. In France, participants used Mon Espace Santé and the private platform Doctolib. Of these systems, only the SAS documentation was accessible for internal analysis through expert interview; the Italian and French systems are analysed exclusively through participant accounts.

In Spain, participants were recruited through two channels: the researcher's personal network and the *Círculo de Escucha* [Listening Circle][1], an informal caregiver support association affiliated with the Hospital Clínico de Málaga. A brief call for participation was distributed through the association's community space, describing the study focus and inviting those who had encountered difficulties navigating health systems on behalf of others to participate. In Italy, recruitment was facilitated through a contact working within a hospital setting, which provided access to participants with direct and recent experience of navigating the SSN as informal caregivers. In France, participants were recruited through the researcher's personal network.

This recruitment asymmetry is acknowledged as a methodological limitation. Institutional recruitment in Spain, through the *Círculo de Escucha*, produced participants with a shared experience of navigating a specific regional health system and a degree of prior engagement with informal caregiver support networks. In Italy, recruitment through a hospital-based contact partially offsets the limitations of personal network recruitment by providing access to participants with direct institutional experience of the SSN as informal caregivers. In France, recruitment relied entirely on personal networks and produced a single complete dyad. This limits the weight that can be placed on the French data independently, though the dyad contributed a methodologically significant observational moment: the real-time configuration of delegated digital access during the interview session itself, which surfaced design failures not detectable through self-report alone. Across all three contexts, the cross-national consistency of the themes that emerged provides partial evidence that the findings reflect structural features of the problem rather than local or recruitment-specific variation.

Inclusion criteria were deliberately broad: participants needed to be either an informal caregiver of a family member with a health condition or a person receiving that care, and to have had direct experience navigating a health system on behalf of or alongside someone else.

[1] The *Círculo de Escucha* is an informal caregiver support group operating within the social services network of the Hospital Clínico Universitario Virgen de la Victoria, Málaga. Participation in the group is open and does not require formal registration.

This breadth was intentional: the goal was not to study a specific condition or demographic but to understand the structural experience of delegation across varied caregiving situations.

The sample was not designed to be statistically representative. It was designed to be conceptually rich. Diversity was sought across caregiving relationship type, health system, digital literacy, age, gender, migration background, and caregiving intensity because these axes of difference shape who bears the burden of informal care and who is most excluded by systems that were not designed with them in mind (Collins & Bilge, 2020; Crenshaw, 1989). One participant, CR08, was recruited at a later stage to address a specific gap: the absence of a caregiving relationship organised outside the heteronormative family model that eHealth systems were built for, in which the informal caregiver's access functions through default institutional assumptions rather than formal recognition.
The original research plan included Sweden as a fieldwork context. Active outreach was conducted in April 2026 through four organisations: Anhörig Skåne, Nationellt kompetenscentrum anhöriga [National Competence Centre for Family Carers], Malmö stad Anhörigkonsulenter [City of Malmö Informal Caregiver Consultants], and the social services division of Malmö stad. Despite confirmed initial interest from Anhörig Skåne, no participants were recruited within the available timeframe. This reorientation was discussed with and supported by the project supervisors.[2]

3.2 Phase 1: Understanding the Current Situation

The primary method was contextual inquiry (Beyer & Holtzblatt, 1998), conducted through semi-structured interviews focused on three areas: the workarounds informal caregivers currently use, the moments where those workarounds break down, and what informal caregivers and care recipients said they actually need from a delegation system. Four interview guides were developed, each adapted to a different participant profile: informal caregiver of an autonomous person, informal caregiver of a non-autonomous person, care recipient, and a combined guide for joint dyadic sessions. All guides focused on the same core areas: everyday workarounds, moments of breakdown or friction, how access and responsibility are negotiated, and implicit and explicit boundaries around health information. Following Flanagan's (1954) critical incident technique, particular attention was given to moments where systems failed, access was denied, or participants had to act without clear authorisation, because those moments reveal the gap between what the system was designed for and what people actually need from it.

Interviews were conducted in Spanish and Italian via Teams, lasting between 20 and 45 minutes. All sessions were recorded with participant consent and transcribed using Microsoft Teams transcription, with subsequent manual verification by the researcher. Transcripts were shared with participants for member checking, allowing them to confirm accuracy, clarify any misrepresentations, and reaffirm or withdraw their continued participation (Lincoln & Guba, 1985). Final consent to include the data in the analysis was confirmed following this review; a small number of participants withdrew at this stage, and their data was excluded entirely. Prior to each session, participants received a data protection document via email, which they returned signed before the interview took place. Informed consent was additionally confirmed verbally at the start of each session. Participants are referred to by

[2] Outreach was conducted via email in April 2026. Anhörig Skåne confirmed initial interest but no participants were available within the project timeframe. Documentation of this correspondence is retained by the researcher

their assigned code throughout the analysis; pronouns are retained where they reflect characteristics central to the analytical argument, particularly in relation to the gendered distribution of caregiving. Data has been handled in accordance with GDPR throughout.

Analysis was conducted using reflexive thematic analysis (Braun & Clarke, 2006), following a six-phase process: familiarisation with the data, generation of initial codes, searching for themes, reviewing themes, defining and naming themes, and producing the report. Affinity diagramming was used alongside thematic analysis to spatially organise and regroup patterns, surfacing design opportunities that emerge from the data rather than from individual accounts. This phase was not designed to produce statistically representative findings but to surface the logic of practice in sufficient depth to inform design. The goal was conceptual saturation: the point at which additional participants would refine rather than fundamentally change the emerging picture.

3.3 Proxy Participation and Ethical Considerations

Three participants in the sample have cognitive or physical conditions that made direct participation impossible; in these cases the person receiving care was not included as a direct participant. This reflects a commitment to ethical integrity, particularly regarding informed consent and the capacity to meaningfully engage in the research process (ACM, 2018). One care recipient participated through asynchronous written exchange over two sessions conducted on separate days to avoid cognitive fatigue, a format chosen in response to the participant's own preference. The data collected through this exchange is treated with the same analytical rigour as interview data, with awareness of its different interactional context.

In these cases, informal caregivers were engaged as proxy informants, asked not only to describe their own practices but also to reflect on the preferences, values, and perceived needs of the person they care for. This is consistent with prior HCI work that frames caregiving as a relational and distributed practice, where interaction with technology is shared across people rather than centred on a single user (Lazar et al., 2018; Storer & Branham, 2023).

3.4 Phase 2: Co-Design Workshops

Phase 2 translates the design opportunities identified in Phase 1 into interaction concepts through participatory co-design workshops. As established in section 2.7, existing approaches to consent and delegation have consistently failed to reflect the relational and contextual dimensions of caregiving (Kaye et al., 2014; Scerpella et al., 2025). Following the logic of Tseng et al. (2024), whose value-centred consent work demonstrates that interfaces designed without patients routinely misalign with patient values, these workshops are not evaluations of a finished design. They are the means through which the design is generated.

Four dyads participated in the workshops, selected to cover the full spectrum from supported use to substituted action (Storer & Branham, 2023) and to maximise variation across delegation type, health system, and caregiving relationship. Table 3 presents the workshop sample. The variation is not incidental: it is what allows the workshops to surface interaction

principles that are robust across different caregiving situations rather than optimised for just one.

Table 3. Workshop sample. Delegation type follows Storer and Branham (2023). Central scenario refers to the scenario used as the primary generative prompt in each session; all dyads were exposed to all three scenarios. Row shading in the Delegation type column follows the colour coding established in Table 2: supported use (green), hybrid (purple), substituted action (red).

Workshop	Dyad	Delegation type	Health system	Language	Central scenario	Rationale
W1	CG08 / CR07	Hybrid	France / Doctolib	Spanish (with French code-switching) [3]	Scenario 1	Both parties have full agency; first configuration of delegation after 15 years
W2	CR05 / CG11	Supported use	Italy / SSN ASL	Italian	Scenario 1	Patient retains control; delegation is functional and segmented
W3	CG04 / CR03	Substituted action	Spain / SAS	Spanish	Scenario 2	CG acts with incomplete information; CR has limited capacity during sessions
W4	CG05 / CR04	Substituted action	Spain / SAS	Spanish	Scenario 3	No contingency network; structural absence of a secondary caregiver

Workshops were structured around three scenarios developed directly from the fieldwork findings, each presenting a concrete moment in the caregiving trajectory: configuring delegation for the first time, acting as a delegate with incomplete information, and managing a contingency when the primary caregiver is unavailable. Participants worked through their central scenario using low-fidelity digital cards that simulated mobile health portal screens without replicating any existing system, keeping the conversation open rather than anchoring it to a specific interface (Figure 6). Each card presented two simultaneous views: what the

[3] CG08 is a Spanish-speaking immigrant in France; CR07 is a native French speaker. The session was conducted in Spanish at CG08's preference. CR07 occasionally used French terms when reading the cards; these were resolved through context and direct clarification during the session.

patient would see and what the informal caregiver would see in the same interaction. This dual perspective was deliberate: it made the asymmetry of access visible within the session itself, provoking negotiation, surfacing silences, and generating the kind of relational friction that single-user interfaces do not expose.

Where participants disagreed about what each view should contain, or fell silent, or asked to change what the other could see, those moments were treated as primary data. The low-fidelity format was deliberate: it invites open feedback rather than closing it down (Buchenau & Suri, 2000). All sessions also included brief exposure to the remaining two scenarios and a structured closing reflection. Blank card templates were provided for generative activities, inviting participants to propose interactions the existing cards did not cover (Figure 7). Sessions were conducted remotely via Teams, lasting approximately 60 minutes.

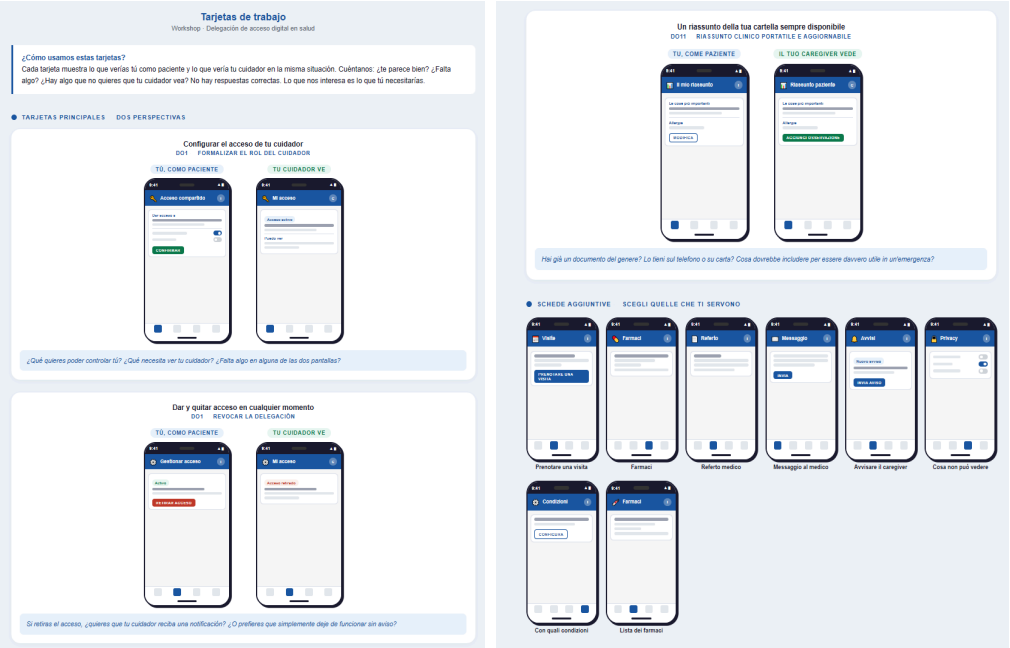


Figure 6. Workshop cards presented to participants during Phase 2 co-design sessions, shown in Spanish (left) and Italian (right). Each card shows the patient view alongside the informal caregiver view of the same interaction, with prompting questions below. Cards were adapted to each participant's language while maintaining the same dual-perspective structure. Cards were designed to simulate mobile health portal screens without replicating any existing system.

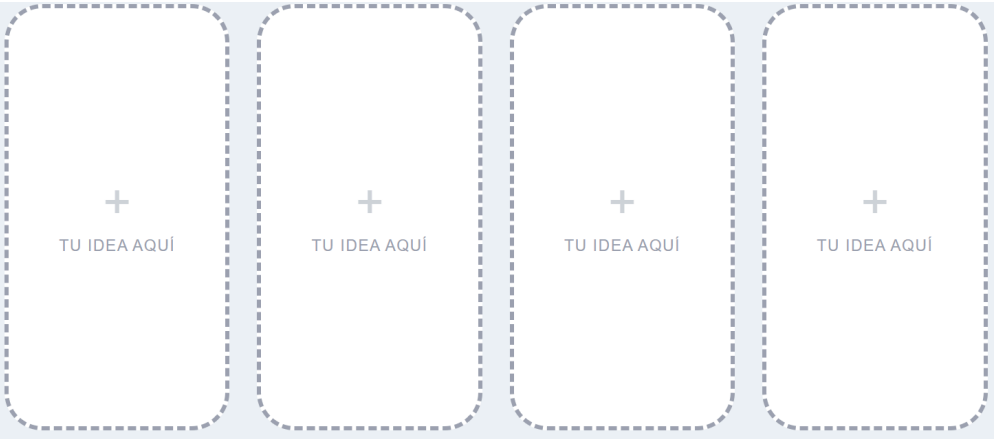


Figure 7. Blank card templates provided to participants for generative activities. The open format was deliberate: participants were invited to propose interactions the existing cards did not cover.

Alongside the co-design workshops, one expert interview was conducted with CG01, a participant with over two decades of experience in the information architecture of the SAS, who also participates in the study as an informal caregiver. This dual role is analytically significant: it produces knowledge that is neither purely institutional nor purely experiential, but situated at the intersection of both. The session was structured as a co-design conversation around the internal documentation of the SAS analysed in section 2.7, and produced evidence about the architectural constraints that shape the systems participants navigate. This material is treated as a distinct source of knowledge that complements the user fieldwork by offering an inside perspective on the design decisions that make informal caregiver invisibility self-reinforcing.

The concepts that emerged from the workshops were developed into three interaction flows covering the core moments of the delegation lifecycle: configuring access, acting as a delegate, and revoking or pausing access. These flows were then developed into a medium-fidelity interactive prototype. Following Houde and Hill's (1997) framework, the prototype focuses on role and look and feel: how delegated access functions in practice and what it feels like to grant, use, and revoke it, rather than on technical implementation. It sits within the visual and interaction language of existing patient portal systems rather than proposing a replacement.

3.5 Phase 3: Evaluating and Reflecting

Phase 3 evaluated the medium-fidelity prototype through individual think-aloud sessions in which participants navigated delegation scenarios while verbalising their reactions, decisions, and moments of uncertainty. Sessions were conducted with five participants across three dyads: CG04 and CR03 (Spain, SAS), CG11 (Italy, SSN), and CG08 and CR07 (France, Doctolib). Each session covered the three interaction flows in full: configuring delegated access, acting as a delegate, and revoking or pausing access. Participants were asked to complete specific actions within each flow, including setting up a configured access, revoking it, and pausing it. This method surfaces not only usability issues but the legibility of the delegation model itself: whether participants understand what access they are granting, to whom, and under what conditions.

Session data was analysed using reflexive thematic analysis (Braun & Clarke, 2006), attending to affective reactions, hesitations, and moments of resistance alongside more explicit responses. Findings feed back into the prototype and into a set of design principles that constitute the project's primary transferable contribution: knowledge about how to design delegated health access given these specific tensions, in these specific contexts, for these specific people.

4. DESIGN PROCESS

This section presents the findings from all three phases of the research process: the thematic analysis of twenty-three interviews conducted in Phase 1, the design principles generated through four co-design workshops in Phase 2, and the interaction flows developed and evaluated in Phase 3. Together they constitute the design knowledge this project produces.

4.1 Overview of Emerging Themes

Twenty-three interviews were analysed using reflexive thematic analysis (Braun & Clarke, 2006), producing ten recurring themes organised around the five perspectives that structure the problem domain: legal and privacy, caregiver, patient, relational and temporal, and design. This structure was not imposed on the data; it emerged from it. The same structural failures appear across all three health systems and all participant profiles, suggesting they reflect a shared design logic that was not built for the people who use it.

Figure 8 maps the full participant sample across the delegation spectrum. The concentration at substituted action reflects the structural absence of formal delegation mechanisms across all three health systems; it is precisely those participants who faced the most acute design failures documented below .

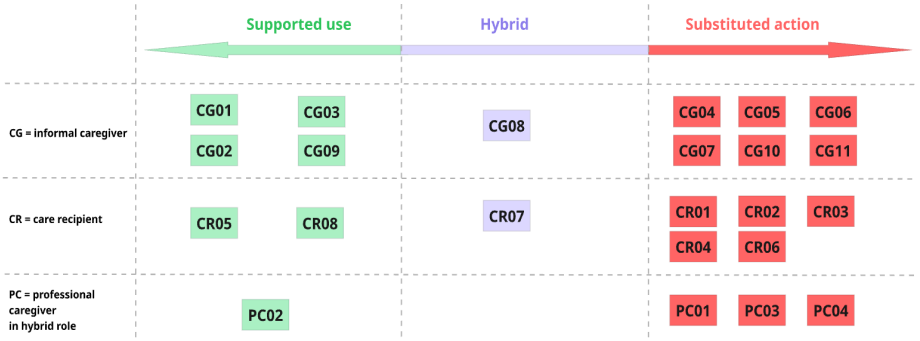


Figure 8. Distribution of the participant sample across the delegation spectrum (Storer & Branham, 2023). N = 23 across Spain, Italy, and France. Rows indicate participant role: CG = informal caregiver; CR = care recipient; PC = professional caregiver in hybrid role. Colour encodes delegation type. The concentration at the substituted action end reflects the structural absence of formal digital delegation mechanisms across all three health systems studied.

4.1.1 Legal and privacy perspective: the gap that forces the workaround

The most consistent finding across the dataset is that informal caregivers perform structural and critical functions within health systems that were not designed to accommodate them.

Every participant described navigating a system built around a binary relationship between the system and the patient, with no formal role for the person acting on that patient's behalf. As one participant put it, the informal caregiver is

> "a bridge: present everywhere in practice, invisible everywhere in design."[4]

That invisibility manifests not as paralysis but as unauthorised necessity. In Spain, CG01 describes how registering his mother's credentials in Cl@ve locked him out of his own account:

> "I had to install Cl@ve with my mother's details, and it removed my own Cl@ve."[5]

A system designed around one identity per device makes delegation technically impossible without a direct cost to the informal caregiver. CG03 manages his mother's private insurance profile using her credentials. CG04 uses her husband's digital certificate with his consent. CG06 managed everything for months using his mother's phone. In Italy, PC02 has her father's profile linked to her own in the regional appointments system. Each of these arrangements involves sharing credentials or accessing accounts that formally belong to someone else, practices that sit outside what these systems were designed to permit. They are not presented here as violations. They are presented as evidence: when a system provides no legitimate channel for something people need to do, they find another way.

The most direct evidence of what this structural absence produces comes from France. During the joint interview with CR07 and CG08, they attempted to configure shared access in Doctolib in real time. Four concrete design failures surface: CG08 accesses CR07's profile without him receiving any notification; access requires only basic personal data that anyone could have; the appointment history she generates is linked to her profile rather than CR07's, fragmenting his clinical record; and at no point is CR07 asked for active consent. CR07 names the implication:

> "With basic data, anyone could access your medical history without you knowing."[3]

This is not self-report. It is a system failing in real time, in front of the researcher.

4.1.2 Informal Caregiver perspective: the workaround as system, no exception

The informal caregiver role emerges organically, without decision or preparation. No participant described choosing to become a caregiver in any formal sense. The role appeared through proximity, capacity, or necessity, and when it appeared, no system explained what support existed or what the informal caregiver was and was not authorised to do.

> CG04: "They give you the diagnosis, and you go home. Nobody teaches you how to deal with any of this."[6]
> CG09: "How did I come to take on this responsibility? Because nobody else in the family did."[7]
> PC04: "There is always one who is the sacrificial lamb."[8]

4 Original: "Tú has sido como un puente entre la tecnología y ellos." (CG01, Spain)

5 Original: "Me tuve que instalar Cl@ve con los datos de mi madre... me quitó mi Cl@ve." (CG01, Spain)

6 Original: "Te sueltan el diagnóstico y te vas a tu casa. Nadie te enseña a lidiar con todo este tipo de cosas." (CG04, Spain)

7 Original: "Come hai fatto ad assumermi questa responsabilità? Perché nessuno della famiglia lo faceva." (CG09, Italy)

8 Originale: "C'è sempre uno che è l'agnello sacrificale." (PC04, Italy)

In the absence of official tools, every informal caregiver in the sample, without exception, has constructed their own documentation system. These are not individual improvisations. They are rational responses to the same structural absence: the official system provides no tools for managing complex clinical information across time. Table 4 maps these systems against their clinical function. Their universality across twenty-three cases, three countries, and very different participant profiles is the most robust finding in the dataset.

Table 4. *Informal documentation systems constructed by participants in the absence of system support, with their clinical function. Each system addresses a gap the official eHealth portal was not designed to fill.*

Participant	Country	Documentation system	Clinical function
CG04	Spain	Handwritten notebook: temperature, time, and duration of each epileptic crisis	Provides the only longitudinal record of symptom patterns between annual neurology appointments
CR04	Spain	Small notebook: every transfusion date, medication change, and treatment update	Functions as the care recipient's only interface with her own clinical history
CG06	Spain	Shared OneDrive folder with siblings; colour-coded calendar for 15 medications with complex dosing schedules	Coordinates care across multiple caregivers at distance; sister used an AI tool to interpret clinical reports
PC03	Italy	Ring-bound notebooks per family member: full clinical history, medication names, doses, and times on the first page	Designed for legibility by a stranger; functions as a portable clinical handover document
CG08	France	Note on phone: allergies, blood type, diagnosis name; physical folder at home organised for any reader to reconstruct the clinical picture	Acts as the only portable clinical summary across hospital systems that do not communicate with each other

The personal cost of this structural invisibility is documented across every caregiver account. CG04 has been diagnosed with caregiver burnout syndrome. CG06 describes the isolation that comes from absorbing a load the system does not see:

> "I felt very bad. The only thing I could do was tell my partner."[9]

9 Original: "Me sentía muy mal. Lo único que podía hacer era contárselo a mi pareja." (CG06, Spain)

That cost is not borne equally. Gender, migration background, and socioeconomic position all shape who absorbs it. CG05 managed three systems simultaneously with no coordination between them: the health system, private insurance, and her mother's legal residency process. She describes a barrier that is not cognitive or technical but relational:

> "We never dared to cross that line of saying, look, give it to me. Never."[10]

Her mother confirms the dynamic from the other side:

> "She worries about bothering people, but we are not bothering anyone. Maybe it is because we are women, I don't know."[11]

4.1.3 Patient perspective: autonomy, ambivalence, and resistance to being displaced

The care recipients interviewed do not form a uniform group. They occupy distinct positions on the delegation spectrum, and each position carries different design requirements. Table 5 maps the four main positions identified in the sample. What they share is a single underlying principle, stated most directly by CR04 and applicable to all of them.

Table 5. Care recipient positions on the delegation spectrum, with their relationship to the system and the design implication each produces. CR08 is discussed separately in the text. Row shading follows the colour coding established in Table 2: supported use (green), hybrid (purple), substituted action (red).

Participant	Delegation position	Relationship to the system	Design implication
CR01	Total delegation, with relief	Cedes all digital health management to CG07 without reservation	System must support full substituted action without requiring the care recipient to remain present in every interaction
CR02	Total delegation, with vulnerability	Delegates everything but has no fallback when primary informal caregiver is unavailable	Delegation without contingency creates structural dependency; secondary access must be designable
CR05	Partial delegation, accepted but not chosen	Delegates by necessity with implicit discomfort; articulates precise criteria: permanent for first-degree family, revocable at will, no renewal	Delegation must feel like a choice, not a surrender; granularity and revocability are not optional features

[10] Original: "Nunca nos hemos atrevido a pasar esa raya de decirle, mire, démela, nunca." (CG05, Spain)
[11] Original: "A ella es que le da apuro molestar, pero no molestamos. Será por ser mujeres, no sé." (CR04, Spain)

CR07	Autonomy with resistance	Actively manages own health, does not identify as a patient, experienced CG08's undisclosed access as a security violation	Delegation configured without the care recipient's active consent is not delegation; it is a privacy failure

The design principle that cuts across all four positions is stated most directly by CR04:

> "Don't push me aside. Let it be clear that she helps me, but that I am still the person."[12]

This is not a preference for a particular interface. It is a statement about identity and agency that any delegation system must take seriously. The resistance to being positioned as passive is not a design obstacle; it is a design requirement.

CR08 adds a position the table cannot fully capture: the care recipient whose delegation functions not because the system recognises the relationship, but because the system has not questioned it. His husband navigates the Italian SSN not through formal recognition of their marriage, but through the system's unquestioned assumption that he is CR08's son. CR08 describes that assumption as the source of his sense of security:

> "Everyone, when we are out and about, everyone takes it for granted that he is my son. They just take it for granted."[13]

That security rests on a systemic assumption rather than a guaranteed right. It dissolves exactly when the system stops assuming and starts asking.

The case of CG10 adds the most extreme dimension in the sample. Presented with a ten-page contract authorising a life-or-death operation, standing up, in full crisis, with a doctor and nurse waiting:

> "If you don't operate, he dies. If you operate, he will have injuries. I did not read the contract."[14]

This is consent as a legal moment without comprehension as a human process. The design implication is direct: consent must be designed as a process, not as a signature.

A related paradox emerges from the comparison between PC03 and CR06. PC03 designed the most sophisticated documentation system in the sample, organised for a stranger to understand. Yet CR06, the care recipient that system was built for, describes it differently:

> "I look at them more to orient myself than to really understand."[15]

The efficiency of the expert informal caregiver can exclude the care recipient even when the information is technically available. Competence, from the care recipient's perspective, can be a form of invisibility.

4.1.4 Relational and temporal perspective: care is not a binary state

12 Original: "Que no me quiten del medio. Que quede claro que ella me ayuda, pero que yo sigo siendo la persona." (CR04, Spain)

13 Originale: "Tutti, quando stiamo in giro, tutti danno per scontato che sia mio figlio. Lo danno per scontato." (CR08, Italy)

14 Originale: "Mi hanno presentato un contratto di quasi dieci pagine. Un medico e un'infermiera mi dicevano: se non operi muore, se operi avrà lesioni. Non ho letto il contratto." (CG10, Italy)

15 Originale: "Aprirei il quaderno... ma io guardo più che altro per orientarmi, non per capire davvero." (CR06, Italy)

CR03 illustrates this with precision:

> "I was used to being very independent, and she tried to help me, but I would not let her."[16]

Delegation did not happen at a moment. It happened gradually, with initial resistance, implicit negotiation, and progressive acceptance. The system has no mechanism for this process. Where delegation mechanisms exist at all, they assume a binary initial state: either you have access or you do not. The process of arriving in that state, which can take months or years, is invisible to the system.

CR05 represents a different position on the same spectrum: delegation accepted but not chosen. He delegates by necessity, with implicit discomfort, and the precision of his design criteria reflects someone who has thought carefully about what he is conceding: permanent authorisation for first-degree family, revocable at will, no renewal. The discomfort is not with the informal caregiver; it is with the absence of a system that would make delegation feel like a choice rather than a surrender.

The French case surfaces the most complex relational dynamic in the sample. CR07 knows he is ill but refuses to accept the extent of his dependence and the corresponding need for formal delegation. He does not know the full scope of what CG08 manages on his behalf. When delegation is configured in Doctolib during the interview, CR07 is surprised to discover CG08 did not already have formal access: CR07 assumed she did; CG08 knew she did not. This asymmetry of knowledge about the state of the delegation is a design failure that no system in the sample addresses. The system did not create this situation, but it did nothing to make it visible, negotiable, or safe.

The most consistent finding across all seven complete dyads points in the same direction. The positive perception of care recipients about the health system does not reflect that the system works well. It reflects that the informal caregiver absorbed all the friction. CR01 describes a calm and positive experience. Her informal caregiver CG07 describes months of solitary learning, a near-serious medication error, and depression. The same situation, two radically different experiences. CR01 could be calm precisely because CG07 absorbed everything she did not have to face.

4.1.5 Design perspective: when the technical solution exists but does not work

PC04 describes an ophthalmology appointment prescribed as urgent but available only in seven months:

> "I already know I will not get it through the regional system, because they give me an appointment in seven months and I don't know if he will still be alive."[17]

France has the most advanced platform in the sample, and the observational evidence from Doctolib demonstrates that even when a technical solution exists, design failures create new problems that did not exist before.

The finding that cuts across all three contexts is not that the systems need more digitalisation. It is that they need better design. CG03 states this with the precision of

16 Original: "Al principio no me dejaba ayudar. Estaba acostumbrado a ser muy independiente." (CR03, Spain)

17 Originale: "So già che la visita oculistica non la farò dal CUP Piemonte, perché mi danno appuntamento a sette mesi e non so se arriva." (PC04, Italy)

someone who understands both sides of the problem:

> "I am a telecommunications engineer and sometimes getting an appointment is hard for me. For an elderly person, it is impossible."[18]

The systems are optimised for the user who needs them least. CG06 names what a different approach would feel like:

> "I don't want systems to seem human. I want them to be easy to understand and to lower my stress."[19]

For an informal caregiver navigating a system that was not designed with them in mind, that is the bar that matters. The following section designs toward it

4.2 Emerging Design Opportunities

The ten themes identified in the previous section converged on a set of structural absences: moments where caregiving relationships required something that eHealth systems were not designed to provide. Eleven design opportunities (DO) emerged from that convergence. Each is grounded in participant evidence and each represents a specific gap between what the delegation relationship requires and what current portals support.

The eleven opportunities are not equal in scope. Five address the structural absence of a formal delegation mechanism directly and were selected for Phase 2: DO1, DO2, DO3, DO4, and DO11. The remaining six describe conditions that surround the delegation relationship, such as clinical language barriers, cognitive load under stress, and institutional coordination failures; they fall outside the scope of the prototype but inform the design principles in section 5. Table 5 maps all eleven opportunities.

Table 5. Eleven design opportunities emerging from Phase 1 fieldwork, mapped against structural absence, participant evidence, design implication, and selection for Phase 2. Row shading alternates for readability only and carries no analytical meaning.

DO	Structural absence	Key participant evidence	Design implication	Phase 2?
DO1	No formal digital role exists for the informal caregiver in any of the three health systems studied.	*CR01, CR06 (Spain, Italy); CG08, CR07 (France)*	Patient-initiated, revocable, multi-caregiver delegation independent of authentication credentials.	Yes
DO2	Caregiver-generated clinical knowledge has no channel into the official health record.	*CG04, CG02 (Spain); PC03, CR06 (Italy); CG08 (France)*	Structured, asynchronous input channel for caregiver observations, reviewable before consultation.	Yes

[18] Original: "Yo soy ingeniero de telecomunicaciones y a mí a veces pedir una cita me cuesta. Para una persona mayor eso es imposible." (CG03, Spain)

[19] Original: "Yo no quiero que los sistemas parezcan humanos. Yo quiero que sean fáciles de entender y que me bajen el estrés." (CG06, Spain)

DO3	Caregivers receive no institutional guidance at the moment the caregiving role begins.	*CG04, CG05, CG06, CG07 (Spain); CR07, CG08 (France)*	Personalised onboarding pathway delivering delegation rights and available mechanisms at point of diagnosis.	Yes
DO4	Systems support one primary caregiver only, with no contingency for their unavailability.	*CG06, CR01, CG04, CG05 (Spain)*	Mechanism for designating primary and secondary caregivers with differentiated access and emergency fallback.	Yes
DO5	Clinical language in portals prevents informed delegation by both parties.	*CG06, CG03, CG01, CG05 (Spain); CG10 (Italy)*	Plain-language summary layer and contextual glossary within delegation interfaces.	No
DO6	Digital channels fail at peak stress moments; human contact resolves what automation cannot.	*CG03, CG05, CG07, CG02 (Spain); CG10 (Italy)*	Visible, low-friction human escalation pathway integrated into delegation flow.	No
DO7	eHealth systems assume a user with full cognitive capacity in conditions of relative calm.	*CG06, CG03, CG05, CG07 (Spain); CG10 (Italy)*	Stress-state design: reduced steps, persistent state, no time-out on delegation tasks.	No
DO8	The caregiver's cumulative load is invisible to the system; no support is surfaced.	*CG04, CG06 (Spain); CG10, CG09 (Italy); PC01, PC02 (Spain)*	Layer that surfaces available caregiver support when substituted action is sustained over time.	No
DO9	Coordination across departments and institutions is performed manually by the caregiver.	*CG08 (France); CG07, CG06, CG05, PC01 (Spain)*	Delegation record legible across departments and institutions, not only within the originating portal.	No

DO10	Caregiver-constructed documentation is sophisticated and clinically significant but invisible to the official record.	*CG07, CG06 (Spain); PC03, CR06 (Italy); PC02 (Spain)*	Structured import pathway for caregiver records into the delegated access layer.	No
DO11	No portable, updatable clinical summary exists; caregivers assemble one by hand.	*CG08 (France); CG07 (Spain); CR07 (France); PC03 (Italy)*	Shared, patient-controlled clinical summary accessible and updatable by the designated caregiver.	Yes

4.2.1 DO1 Formalising the Caregiver's Digital Role

Every case in the sample involves an informal caregiver performing digital health management without any legitimate mechanism to do so. The most direct evidence comes from France. During the joint interview with CR07 and CG08, they attempted to configure shared access in Doctolib in real time. Four concrete design failures surfaced: CG08 accessed CR07's profile without him receiving any notification; access required only basic personal data that anyone could have; the appointment history she generated was linked to her profile rather than CR07's, fragmenting his clinical record; and at no point was CR07 asked for active consent. CR07 names the implication:

> "With basic data, anyone could access your medical history without you knowing."[20]

CR08's case adds a dimension that standard access criteria do not capture. His husband navigates the Italian SSN through the system's unquestioned assumption that he is CR08's son. That assumption dissolves precisely when it is needed most: in an emergency, in a complex medical decision, in any context where the system stops assuming and starts asking. The only criterion that works across the full diversity of caregiving relationships in this sample is the patient's active and explicit consent.

The design criteria that emerge consistently across Spain, Italy, and France: delegation should be revocable at any point by the care recipient, compatible with multiple informal caregivers in differentiated roles, and independent of the care recipient's own authentication credentials. DO1 is the foundation on which all other design opportunities rest.

4.2.2 DO2 A Channel for Informal Caregiver-Generated Clinical Knowledge

A recognised digital role for the informal caregiver is not sufficient on its own. Between consultations, which in many cases happen once a year, informal caregivers accumulate continuous, clinically relevant knowledge about the person they care for. That knowledge has no official channel into the health record. CG04 documents every epileptic crisis in a notebook:

[20] Original: "Con unos datos básicos cualquiera podría acceder a tu historial sin que tú lo supieras." (CR07, France)

> "I write everything down: the temperature, the time, how long it lasts. If you don't write it down, you only see the neurologist once a year, and the other 364 days we're alone at home."[21]

CG02 prevented an unnecessary procedure because she knew her aunt had a condition that did not appear anywhere in the medical record the doctor was consulting. CG08 keeps a note on her phone with her husband's allergies, blood type, and diagnosis name, assembled entirely by hand because no system provides one. Following Hutchins's (1995) concept of distributed cognition, caregiving already functions as a distributed clinical observation system. The problem is that the health system has no channel to receive what that system produces. DO2 is the informational layer that makes DO1 meaningful: a formal role without a contribution channel produces a caregiver who can see the record but cannot add to it.

4.2.3 DO3 Institutional Onboarding After Diagnosis

Knowing that a delegation option exists is a precondition for using it. Yet every participant described learning to navigate the health system through trial, error, and informal networks, with no structured guidance at the moment the caregiving role began. The data reveals three structurally distinct forms of institutional abandonment at the point of delegation, each producing a different kind of failure: not knowing that delegation exists as a possibility; not knowing what one is authorised to do; and not knowing how to configure access when it does exist. Figure 9 maps these three forms and their evidence across the sample.

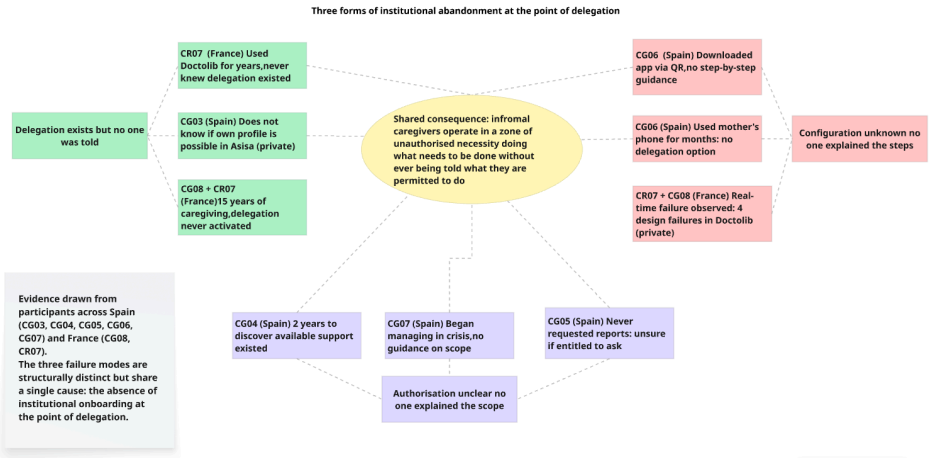


Figure 9. Three forms of institutional abandonment at the point of delegation, with participant evidence across Spain and France. The three failure modes are structurally distinct but share a single cause: the absence of institutional onboarding at the moment when the caregiving relationship formally begins. Evidence drawn from CG03, CG04, CG05, CG06, CG07 (Spain) and CG08, CR07 (France).

CR07 illustrates the first form, discovering during the interview that a delegation function in Doctolib exists and has never been activated:

> "I suppose there must be some means by which I can delegate access to someone. I need to look into that. I don't use it. I don't have it."[22]

[21] Original: "Yo lo apunto todo: la temperatura, la hora, lo que dura. Como no lo apuntes, te ve una vez al año al neurólogo y los otros 364 días que estamos solos en casa." (CG04, Spain)
[22] Original: "Ahí supongo que sí debe haber algún medio que yo dé delegación a alguien para que tenga acceso a eso. Lo tengo que mirar. No la uso. No la tengo." (CR07, France)

CG04 illustrates the second, spending two years before discovering she was entitled to request formal institutional support:

> "It took me about two years to find out I could request help for my husband."[23]

CG05 never requests clinical reports because she does not know whether she has the right:

> "We never dared to cross that line of saying, look, give it to me. Never."[24]

CG06 illustrates the third, downloading the regional health app by following a QR code on a hospital wall with no guidance:

> "They don't explain how the app works. You go to the QR, download it, and you have to start with the certificate, the ID and goodness knows what else. A mess."[25]

4.2.4 DO4 Distributed Caregiving with Contingency

DO1 addresses the absence of a formal caregiver role. DO4 addresses what happens when that role is held by only one person. Every case in the sample involves a primary informal caregiver who concentrates all management responsibility, not by choice but as the structural outcome of a system that grants access to only one person and has no mechanism for distributing that role.

CG06 had arranged for his sister to take over when he left. His father fell ill in the interval:

> "There was a moment when my father got ill, and it was a mess because there was no way to manage it."[26]

CR01 describes the same failure from the other side: when CG07 was unavailable, there was no one else. "Who took care of it? I did it myself."[27] The fear of one's own unavailability runs across the sample. CG06 names the structural impossibility directly:

> "When you are taking care of all of this in your family, you cannot get ill yourself."[28]

DO4 is a specific interaction design problem: the system currently has no mechanism for designating primary and secondary informal caregivers with differentiated access, and no way for a secondary informal caregiver to step in without starting from scratch.

4.2.5 DO11 A Portable and Updatable Clinical Summary

The four design opportunities above address the relational structure of delegation: who has access, how it is granted, how it is communicated, and how it is distributed. DO11 addresses the informational infrastructure that makes all of them functional. Without a shared clinical

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[23] Original: "Fue como dos años en saber que podía pedirles ayuda para mi marido." (CG04, Spain)
[24] Original: "Nunca nos hemos atrevido a pasar esa raya de decirle, mire, démela, nunca." (CG05, Spain)
[25] Original: "No te explican cómo funciona la aplicación. Tú te vas al QR, te la descargas y tienes que empezar con el certificado, con el DNI y no sé cuánto. Un follón. No te explican cómo tienes que hacerlo poco a poco." (CG06, Spain)
[26] Original: "Después hubo un momento en el que mi padre, cuando se estaba encargando después de que yo me fuese y en el intervalo antes de que llegase mi hermana, se pone enfermo y eso fue un follón porque no había forma de gestionarlo." (CG06, Spain)
[27] Original: "¿Quién se encargó? Pues lo tuve que hacer yo." (CR01, Spain)
[28] Original: "Porque cuando tú te estás encargando de todo esto en tu familia, tú no te puedes poner enfermo." (CG06, Spain)

summary, an informal caregiver with delegated access can see the record but cannot orient themselves within it.

Every informal caregiver in this sample has built a version of the same artefact: a portable clinical summary assembled by hand. CG08 keeps a note on her phone:

> "In my phone I always have the document with the allergy, with his blood type, with what his illness is called."[29]

When CR07 was hospitalized, CG08 acted as the only link between clinical teams with no shared record, carrying documents physically between hospitals. CR07 has done the intellectual work of identifying which documents matter, but when asked whether they are organised so that someone else could find them, his answer is immediate: "No, no, no."[30] CG08 names what fragmentation costs:

> "Every time you go to a new doctor you have to start from scratch."[31]

DO11 is not a delegation mechanism. It is what makes delegation meaningful. Without a shared clinical summary, the other four design opportunities solve the structure of access without solving what access is for.

4.3 Co-Design Workshops: From Design Opportunities to Design Principles

The five primary design opportunities translated the three sub-questions into concrete design territory. SQ1 had been answered analytically in Phase 1; what remained was to understand what those failures required from an interaction design perspective. SQ2 could not be answered through fieldwork alone: it required putting design proposals in front of people and observing how they negotiated them. SQ3 was the most directly generative question for Phase 2. Four co-design workshops were conducted to answer all three, each bringing together one informal caregiver and one care recipient who had participated in Phase 1. Table 6 maps the four workshops against their participant profiles and primary contribution to each research question.

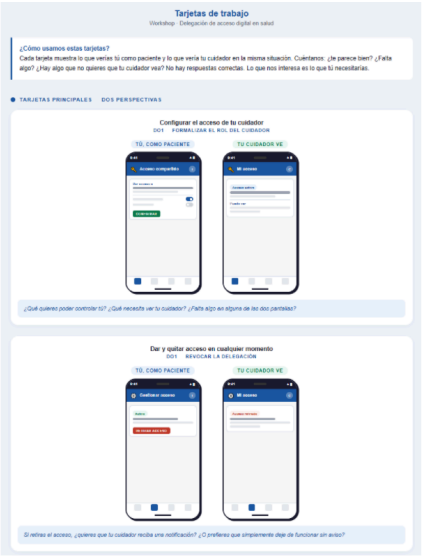


Figure 10. Dual-perspective card format used in the workshops. Each screen card presents the patient view and the caregiver view simultaneously. Guided questions below each card prompted negotiation rather than individual evaluation.

The material consisted of eighteen screen cards grouped by scenario and delegation moment, with eight additional cards introduced only if participants identified needs the main cards did not address. Several of the most significant findings emerged precisely from these secondary and blank cards. The researcher sketched proposed changes in real time in Figma and checked with participants whether the sketch captured what they had described. Figure 10 shows an example of real-time sketching from Workshop 3.

[29] Original: "Yo por ejemplo tengo una nota hecha en mi teléfono con todas las alergias, el tipo de sangre, la enfermedad, cómo se llama su enfermedad." (CG08, France)
[30] Original: "No, no, no." (CR07, France)
[31] Original: "Cada vez que vas a un médico nuevo tienes que empezar de cero. Su servicio médico actualmente no tiene eso." (CG08, France)

Participation patterns varied within and across sessions. The most contested moments, typically around who sees results first or what counts as urgent information, were treated as primary analytical material. Silences were equally informative: the screens that produced the longest pauses or explicit disagreement surfaced the most significant design requirements. Several participants noted, unprompted, that the workshop had been the first time they had discussed delegation directly with each other, even in cases where delegation had already been happening for years. In all four sessions the informal caregiver contributed more extensively to the design discussion than the care recipient, reflecting the structural position of the person who navigates and fails against the system most. The care recipient's contribution, though less extensive, consistently anchored the discussion in the question of agency, preventing caregiver-centred solutions from going unchallenged. This asymmetry is itself a finding about SQ2 and shaped the design principles in Clusters II and IV directly. Table 6 maps the four workshops against their participant profiles, delegation type, and primary contribution to the research questions.

Table 6. *Overview of the four co-design workshops: participant pairs, national context, delegation type, DOs activated, and primary contribution to each research question. Row shading in the Delegation type column follows the colour coding established in Table 2: supported use (green), hybrid (purple), substituted action (red).*

WS	Pair	Country	Delegation type	DOs activated	Key contribution
W1	*CG08 / CR07*	*France*	*Hybrid*	*DO1, DO2, DO3, DO4, DO11*	*8 new principles: two-layer access, active audit, two-author summary, AI boundary. Confirmed the delegation conversation had not occurred in 15 years. Primary source for SQ3.*
W2	*CR05 / CG11*	*Italy*	*Supported use*	*DO1, DO2, DO5, DO11*	*Temporal granularity; patient as emotional intermediary. Most explicit contradictions in the sample, refining several principles from universal to context-specific. Primary source for SQ2.*

W3	*CG04 / CR03*	*Spain*	*Substituted action*	*DO1, DO2, DO3, DO4, DO9, DO11*	*Analogue documentation not replaced but connected. Dignity as design constraint. Strongest confirmation of DP-19. Primary source for SQ1.*
W4	*CG05 / CR04*	*Spain*	*Substituted action*	*DO1, DO3, DO4, DO11*	*Pro tempore permission as partial response to absent contingency networks. DP-25 confirmed as system limit. Multilinguality requirement for clinical summary. Primary source for SQ1 and SQ2 in migrant caregiving context.*

4.3.1 Workshop 1 CG08 and CR07 (France, hybrid delegation)

The first workshop was the primary source for SQ3 and produced the most analytically rich material of the four sessions. The session opened with a discovery that set the analytical frame: CR07 had assumed CG08 already had formal access to his portal; CG08 knew she did not. The delegation conversation had not occurred in fifteen years of caregiving, not because the pair lacked the will but because no system had created the conditions for it to happen.[32]

The most precise formulation of gradual delegation across all four workshops came from CR07:

> "First me. Then her if I decide to tell her, or if I can no longer manage it alone."[33]

This is not a binary access model but a sequence with two distinct activation conditions: active decision and incapacity. Each requires different design mechanisms.

CG08 named what onboarding should actually do:

> "Onboarding should not be configuring an access. It should be having this conversation. And the system should facilitate it. With questions. With time. Without rushing."[34]

[32] Original: "No me lo dijiste." / "No querías hablar de ello. Estabas mal. Y cuando estás bien no quieres hablar de cuando estás mal." (CR07 / CG08, Workshop 1, France)
[33] Original: "Primero yo. Luego ella, sí yo decido contárselo, o si no puedo gestionarlo solo." (CR07, Workshop 1, France)
[34] Original: "El onboarding no debería ser configurar un acceso. Debería ser tener esta conversación. Y el sistema debería facilitar lo. Con preguntas. Con tiempo. Sin prisa." (CG08, Workshop 1, France)

CR07 articulated the relationship between revocability and trust:

> "If I know I can take it back, I can give more."[35]

Reversibility here is not primarily a safety mechanism; it is what makes a broader scope of delegation feel possible. This is the empirical basis for DP-15.The session produced eight principles not present in Phase 1, including the two-layer access architecture, the active audit function, the two-author clinical summary, and the boundary principle for LLMs in clinical contexts: simplify language and flag value changes yes; interpret or diagnose no.

4.3.2 Workshop 2 CR05 and CG11 (Italy, supported use)

The second workshop was the primary source for SQ2 and produced the most explicit contradictions in the sample. CR05 retains full digital agency and delegates selectively. His most significant contribution was a spontaneous distinction between recent and historical diagnoses: he wanted to filter recent ones before sharing them with his informal caregiver, while historical diagnoses could be shared freely. The criterion was proximity in time, not content, and it points to a configurable access category no system in the sample currently offers.

CR05 also introduced the patient as an emotional intermediary: the system should defer the informal caregiver's access to a serious diagnosis until the patient has reviewed it. "If they tell you there is cancer, you don't want the person who loves you to be hit with it without warning."[36] This is not a privacy restriction; it is a relational design requirement confirmed from a complementary angle in Workshop 4 and documented as context-specific pending broader testing.

Workshop 2 also produced the most important methodological constraints of the four sessions. The audit log, automatic access expiry, and constant notifications were all rejected here with direct participant evidence, preventing several principles from being generalised beyond the profiles that produced them.

4.3.3 Workshop 3 CG04 and CR03 (Spain, substituted action)

The third workshop was the primary source for SQ1. CG04 manages CR03's health without any formal digital mechanism and maintains the most clinically detailed handwritten documentation in the study. Her response to the scenario proposing a structured digital log was immediate:

> "I will keep the notebook regardless of what the app offers. What I want is for what I write to reach the doctor."[37]

This is the clearest formulation of DO2 from the informal caregiver's perspective: the goal is not to replace the analogue system but to give it a clinical channel. The session surfaced dignity as a design constraint the previous material had not named directly. CR03 resists being positioned as a patient who needs an informal caregiver; his identity is organised around autonomy. When the onboarding scenario asked him to categorise himself as a

[35] Original: "Si sé que puedo quitárselo, puedo dar más." (CR07, Workshop 1, France)
[36] Original: "Se ti dicono che c'è il cancro non vuoi che una persona che ti vuole bene venga assalita da questo." (CR05, Workshop 2, Italy)
[37] Original: "Mantendré la libreta independientemente de lo que ofrezca la app. Lo que quiero es que lo que escribo llegue al médico." (CG04, Workshop 3, Spain)

person requiring assistance, he rejected it on identity grounds. CG04 named the implication: "Don't put a caregiver. I am his wife."[38] Combined with CR07's equivalent rejection in Workshop 1, this established DP-03 as a nuclear principle: trust in the system requires that the system sees the relationship as the participants see it.

Workshop 3 also introduced two principles outside the scope of the three interaction flows that structure the prototype: a three-level communication channel and a system-generated urgency signal. Both require institutional architecture beyond the prototype. The urgency signal was proposed precisely because the alternative, a button asking the informal caregiver to label something as urgent, was explicitly rejected in this and two other workshops. Figure 11 shows the real-time wireframe produced during the session, including the traffic-light system proposed by CG04 to represent clinical urgency.

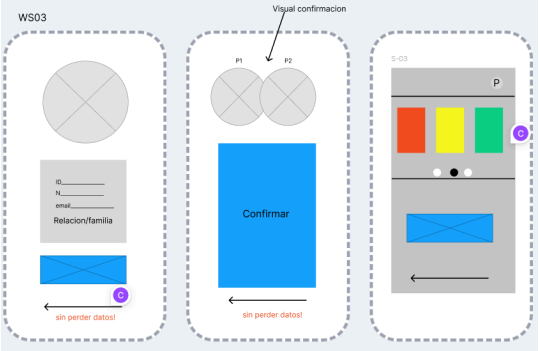


Figure 11. Real-time wireframe produced during Workshop 3. The traffic-light system (red / yellow / green) on screen S-03 was proposed by CG04 to represent clinical urgency without requiring the caregiver to interpret it as the basis for DP-23. *"Sin perder datos"* (Without losing data) appears on two screens, capturing verbatim a participant requirement: navigating back must not cause data loss. *"Relación/familia"* (Relationship/family) was added after participants indicated the existing categories did not reflect their relationship.

4.3.4 Workshop 4 CG05 and CR04 (Spain, substituted action, migrant caregiving context)

The fourth workshop made visible what the other three could not: the situation of informal caregivers without a contingency network. CG05 manages her mother's health without extended family, with limited institutional support, and with a legal residency process running in parallel. When the scenario introduced the concept of a secondary informal caregiver as a contingency mechanism, her response named the limit of the design:

> "We have nobody. That is the problem. I have nothing to configure."[39]

The pro tempore permission (DP-11), a time-limited access granted for a single action without requiring a permanent informal caregiver profile, is a partial solution for low-stakes logistical actions. It does not resolve the structural situation of an informal caregiver who is the only person available. This is DP-25: the point where the design argument reaches its boundary and institutional intervention begins.

CR04 contributed the most direct formulation of DP-13: both parties must see the consequences of delegation before either accepts. The confirmation step is not the beginning

[38] Original: "No quiero que pongan cuidador. Yo soy su mujer." (CG04, Workshop 3, Spain)
[39] Original: "Nosotros no tenemos a nadie. Ese es el problema, que no tengo nada que configurar." (CG05, Workshop 4, Spain)

of the consent process; it is its conclusion. This directly answers SQ3. Workshop 4 also introduced the multilinguality requirement for the portable clinical summary: for participants living across national health systems, the summary must be available in a language the receiving clinical team can use, extending DO11 beyond its original scope.

4.3.5 Synthesis: What the Workshops Produced and What They Could Not

Of the twenty-five principles, fifteen are nuclear, holding across participant profiles, delegation types, and national contexts. Four are context-specific. Two were contradicted and retained as options. Three exceeded the scope of the prototype, and two reached the boundary of what interface design can resolve. The workshops were not a validation exercise: they produced new requirements, introduced contradictions that refined universal principles into context-specific ones, and documented explicit rejections that are as analytically valuable as the confirmations. Table 8 maps the full distribution.

Table 7. *Three synthesis operations applied to the raw workshop material to produce the design principles.* Row shading alternates for readability only and carries no analytical meaning

Operation	Definition	Result in this study
Collapse	Principles describing the same requirement from different sources were merged into one.	25 principles from a larger initial set of observations across 4 workshops and the expert interview.
Elevation	Principles confirmed in 3 or more independent sources were marked as nuclear.	15 nuclear principles, transversal across participant profiles, delegation types, and national contexts.
Preservation	Principles with a single source covering territory no other principle addressed were retained as context-specific.	4 context-specific principles, 2 contradicted principles retained as options, 3 out of scope, 2 system limits.

Table 8. *Twenty-five design principles synthesised from the four workshops and the expert interview, organised by cluster. Status codes: Nuclear = 3+ independent sources; Confirmed, localised = 2 direct sources; Context-specific = 1-2 sources; Contradicted, option only = valid for some profiles; Out of scope = exceeds prototype; System limit = cannot be resolved by interface design. Row shading indicates principle status: nuclear (blue), confirmed and localised (teal), context-specific (amber), contradicted or option only (yellow), out of scope (light grey), system limit (grey).*

DP	Principle	Cluster	RQ	Sources	Status
DP-01	Caregiver has a distinct session, action log, and authenticated identity, separate from the patient's	I. Identity	SQ1	Phase 1, W1-W4, CG01	Nuclear confirmed all sources

DP-02	Clinical team is informed when an action was performed by the informal caregiver, not the patient	I. Identity	SQ1	Phase 1, W4, CG01	Nuclear
DP-03	Interface language does not impose clinical identities users have not adopted ("trusted person", not "caregiver")	I. Identity	SQ3	W1-W4	Nuclear 4 workshops, both roles
DP-04	Delegation is configurable before crisis, not only after	II. Config	SQ2	Phase 1, W1, W2, W4, CG01	Nuclear
DP-05	Delegation is granular by functional category: each category toggled independently	II. Config	SQ2	Phase 1, W2, W3	Nuclear
DP-06	Access granularity includes a temporal dimension: recent vs. historical diagnoses configurable separately	II. Config	SQ2	W2 (CR05)	Context-specific confirm before generalising
DP-07	Patient can act as emotional intermediary: defers informal caregiver access to high-impact news until patient has reviewed it	II. Config	SQ2	W2 (CR05), W4 (CR04)	Context-specific two sources, complementary logics
DP-08	System offers at least three delegation models as starting points without imposing any	II. Config	SQ2, SQ3	W1, W2, W3	Nuclear
DP-09	Emergency access and everyday access are distinct controls that cannot be	II. Config	SQ1, SQ2	W1, W3	Confirmed, localised

	linked or revoked together				
DP-10	Cascaded delegation requires explicit rules set by the patient; primary informal caregiver cannot extend access unilaterally	II. Config	SQ2	W2, W3	Confirmed, localised
DP-11	Pro tempore permission available for one-off actions by persons without a permanent informal caregiver profile	II. Config	SQ2	W4	Context-specific partial solution for absent contingency networks
DP-12	Onboarding is a facilitated conversation between CR and CG, not a configuration form	III. Onboarding	SQ3	W1-W4, CG01	Nuclear strongest confirmation in the corpus; both roles in all workshops
DP-13	Both parties see the consequences of delegation before either accepts	III. Onboarding	SQ3	W4, Phase 1 (CG10)	Nuclear
DP-14	Patient receives notification when an informal caregiver is added; this is non-negotiable	IV. Security	SQ3	Phase 1, W1-W4, CG01	Nuclear
DP-15	Patient can revoke informal caregiver access at any time, without friction; technically trivial per CG01	IV. Security	SQ2, SQ3	W1-W4, CG01	Nuclear
DP-16	Any action on third-party health data requires second-factor authentication	IV. Security	SQ3	F1, W1-W4, CG01	Nuclear
DP-17	One authentication gives access to all	IV. Security	SQ3	Phase 1, CG01	Nuclear architectural constraint

	delegated functions (SSO)				confirmed by expert
DP-18	Patient can view a record of informal caregiver activity; offered as opt-in, not default	IV. Security	SQ2, SQ3	W1, W2, W4	Contradicted, option only CR07 requests it; CR05, CG11, CR04 reject it as default
DP-19	Informal caregiver is a structured clinical source: their observations reach the clinical team, not just the portal	V. CG as source	SQ1	W1-W4	Nuclear transversal; strongest confirmation in W3
DP-20	AI simplifies clinical language and flags value changes; does not interpret or diagnose	V. CG as source	SQ1, SQ3	W1, W2, W4	Nuclear autonomous interpretation explicitly rejected in 3 workshops
DP-21	Portable clinical summary has two authors: patient data and caregiver observations as co-equal inputs	VI. Summary	SQ1, SQ2	W1-W4	Nuclear transversal
DP-22	Three-level communication channel: urgent / semi-urgent / deferred messages to clinical team	VI. Summary	SQ1	W3 (CG04)	Out of scope requires institutional architecture
DP-23	System-generated urgency signal based on clinical data model, not caregiver interpretation	VI. Summary	SQ1	W3, W4	Out of scope urgent button rejected in 3 workshops; signal requires clinical data integration
DP-24	Delegation threshold must not require a formal disability grade	System limit	SQ1	F1, W1-W4, CG01	System limit current threshold excludes the entire sample

DP-25	System cannot assume a contingency network exists	System limit	SQ2	W4 (CG05)	System limit DP-11 is partial response only

One source outside the workshops is critical to the integrity of Cluster IV. CG01 holds a double analytical position in this study: he is the information architect of the Andalusian Health System portal, and he is also an informal caregiver. His testimony operates on two layers: technical confirmation of what the system can and cannot do today, and experiential confirmation of what it costs when it does not do it. His confirmation that revocation is technically trivial and absent by design omission transforms DP-15 from a speculative proposal into a documented implementation gap. The same expert confirmed that the absence of second-factor authentication for third-party data access is a known vulnerability and that the credential-sharing workaround documented across the entire Phase 1 sample is its direct consequence.[40]

One cross-cutting finding shaped the flows without appearing as a discrete principle in any single workshop: the moments of greatest need for delegation coincide, systematically, with the moments of lowest capacity for interaction. CG10 signed a ten-page consent form standing up, in full crisis. CG08 asked what happens when she is in an emergency. CG06 named the structural impossibility of getting ill while being the only person available. DP-09, which separates emergency from everyday access, operationalises this observation directly. The body of the informal caregiver is not a contextual variable. It is a design constraint.

The twenty-five principles form a system organised around a single argument: delegation is not a technical state to be configured; it is a relationship to be supported over time. The clusters move from identity to configuration to consent to security to clinical contribution, following the arc of a delegation relationship from its first moment to its ongoing practice. The two system limits document where that arc reaches its boundary. What the prototype must now demonstrate is whether these principles, when materialised as interaction flows, produce something a person navigating a real caregiving relationship would actually use.

5. MAIN RESULTS AND FINAL DESIGN

The three-phase study documented sharing credentials, using a patient's device, and attending appointments in person to access information that should be reachable digitally as unauthorised necessity. This chapter presents them as a design problem with a designable

[40] Four design elements were explicitly rejected across three or more workshops. (1) LLMs as autonomous clinical interpreters: CG08 (W1) rejects clinical synthesis by LLMs; CR05 (W2): "purché qualcuno prenda responsabilità" ["as long as someone takes responsibility"]; CR04 (W4) accepts LLMs for language translation only. Rejected in three workshops; the accepted distinction is: simplify language and flag changes yes, interpret or diagnose no. (2) The "urgent" button as a caregiver label: CG04 (W3): "no quiero tener que preguntar, el sistema debe decirme directamente el nivel de importancia" ["I don't want to have to ask; the system should tell me directly how important it is"]; CR04 (W4): "a mí no me gusta que la aplicación le haga al cuidador decidir si esto es urgente" ["I don't like the app making the caregiver decide whether something is urgent"]; CG11 (W2) prefers a notification, not a button. Rejected in three workshops. (3) The audit log as a default visible function: CR05 (W2): "probabilmente gli toglierei anche la responsabilità da questo punto di vista" ["I would probably remove that responsibility from them too"]; CG11 (W2) does not want a record of their own actions; CR04 (W4): "el historial de lo que tú has visto... eso me da absolutamente igual" ["the record of what you have seen... that is completely irrelevant to me"]. Rejected in two workshops; retained as opt-in following CR07's contrasting position. (4) Automatic expiry of the primary caregiver's access: CG08 (W1): "que te dejes la libertad de que, cuando quieras eliminarlo, lo eliminas y punto" ["just leave yourself the freedom to delete it whenever you want, and that is it"]; CR05 (W2): "durata dell'accesso non lo vedo necessario" ["I don't see a time limit on access as necessary"]. Rejected in two workshops for permanent access; accepted in W4 for the pro tempore permission only.

answer. The three interaction flows, the medium-fidelity prototype, and the evidence from the think-aloud evaluation together constitute that answer. They do not resolve the tensions between autonomy, access, and care that the fieldwork surfaced; they make the designable ones navigable.

5.1 The Final Design

The prototype covers the delegation lifecycle across three interdependent flows. Flow 1 addresses the moment of configuration: who is recognised, what is shared, and under what conditions (SQ3). Flow 2 addresses ongoing negotiation: how delegation is adjusted, paused, or revoked as circumstances change (SQ2). Flow 3 addresses the exercise of delegated access: how an informal caregiver acts within the system and contributes information back to the clinical record (SQ1). The flows are not modular. An informal caregiver cannot act in Flow 3 without a delegation configured in Flow 1, and a delegation configured in Flow 1 is only meaningful if it can evolve through Flow 2.

The prototype sits within the visual and interaction language of existing patient portal systems rather than proposing a replacement. Following Houde and Hill's (1997) framework, it focuses on role and look and feel: what delegated access means in practice and what it feels like to grant, use, and revoke it, rather than on technical implementation. Each flow is described in detail in section 5.2, with selected prototype screens and interaction flow maps illustrating the key interaction moments. Full prototype screens are included in Appendix A .

5.2 What the Design Does

Flow 1: configuring delegation. The calm-state sub-flow (1a) begins at the patient dashboard, where a persistent status indicator surfaces delegation as something the patient can see and change at any moment, not a buried settings item. This operationalises DP-01 and DP-14 simultaneously: the informal caregiver has a recognised role, and the patient is notified whenever that role changes state.

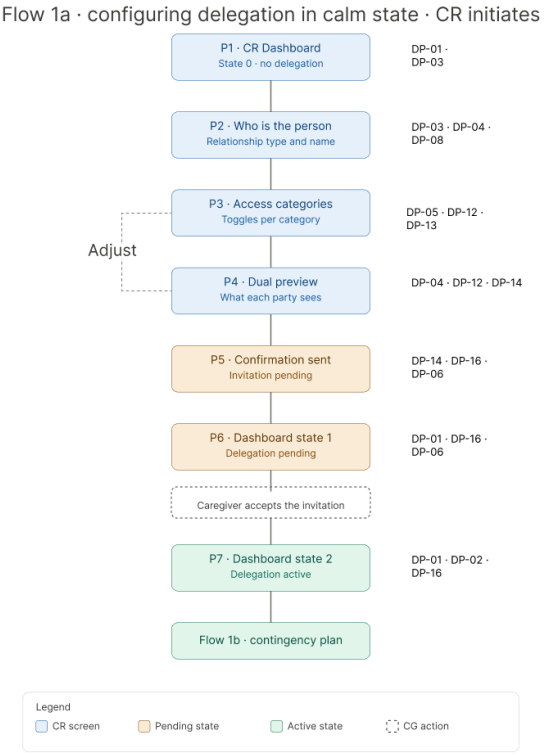


Figure 12. Interaction flow map for Flow 1a (configuring delegation in calm state). The flow begins at the patient dashboard with no delegation configured and ends with delegation active. Each screen references the design principles it operationalises. CR = care recipient screen; CG action = action performed by the informal caregiver.

The patient identifies the informal caregiver through a relational field that accepts any combination of proper name and relational label rather than a clinical category. This implements DP-03 directly. The evidence for this principle runs through every phase of the project: CG04 rejected *"caregiver"* as the label for her husband in Workshop 3 and all three think-aloud participant groups used their own relational language without prompting. The field is the design's response to the finding that institutional categories can make a care relationship feel foreign to the people living it.

The permission screen presents functional categories (appointments, active medication, results, prior diagnoses) as independent toggles, each starting disabled. The patient builds the agreement actively rather than dismantling a pre-configured one. A confirmation screen then shows both views simultaneously: what the CR is granting and what the informal caregiver will see. This dual-perspective design, developed in Phase 2 from the co-design cards that showed both views in parallel, implements DP-13. CG08 described this screen in the think-aloud as "very complete",[41] and for CG08 and CR07 it was the first moment in fifteen years of caregiving in which they had discussed the scope of digital health access directly.

[41] *Original: "Vista de Carlos, vista tuya, muy bien, muy completa."* (CG08, Think-aloud Session 1, France)

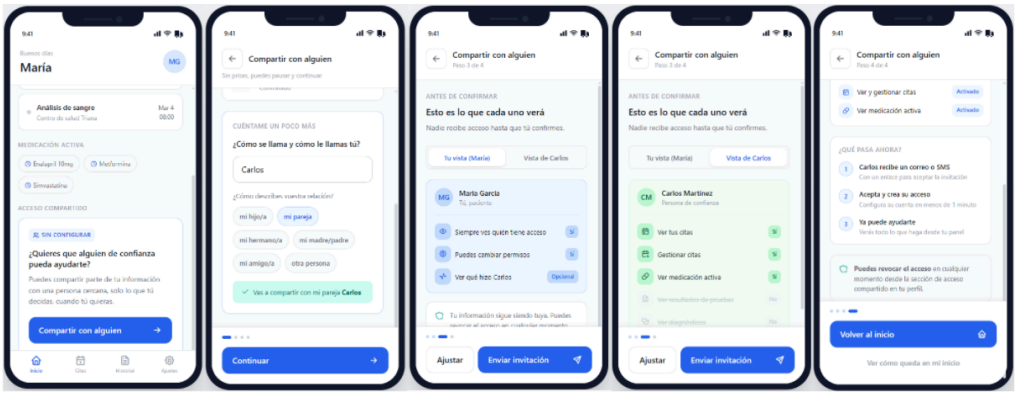
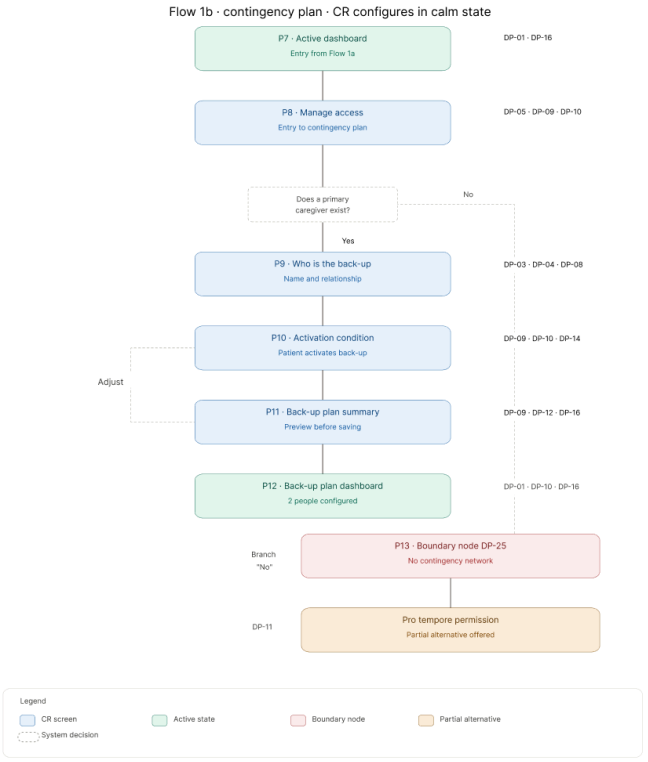


Figure12. Selected screens from Flow 1a (configuring delegation). From left to right: the delegation entry point, showing the persistent status indicator on the patient dashboard (DP-01, DP-14); the relational identification field, which accepts any name and relationship label the patient chooses (DP-03); the dual-perspective confirmation screen showing simultaneously what the patient and the informal caregiver will see before either confirms (DP-13); and the post-configuration screen showing the revocation option as a persistent and accessible control (DP-15).

The contingency sub-flow (1b) introduces the design decision that required the most significant revision between Phase 2 and Phase 3. The think-aloud evaluation revealed a critical failure in the original flow: participants could not tell whose perspective they were operating from when configuring a back-up informal caregiver. They assumed they were still acting as the care recipient when the interface had silently shifted to the primary informal caregiver's view. CR07 named the failure precisely:

> "I saw this screen as if I were Maria. And I was adding another person as a backup."[42]



[42] Original: *"Esta pantalla yo la veía como si fuese yo María. Y estaba añadiendo otra persona de respaldo."* (CR07, Think-aloud Session 1, France)

Figure 13. Interaction flow map for Flow 1b (contingency plan). The flow branches at the question of whether a primary caregiver exists. The boundary node P13 documents DP-25: the system limit reached when no contingency network is available.

The revised design introduces an explicit transition screen before the back-up configuration begins (Figure 13). The screen addresses the care recipient directly and makes the control logic visible before asking for any configuration: the back-up contact only becomes active when the patient decides, and that person must accept before having access. This screen is the clearest instance in this project of Research through Design producing a design response to an empirical finding: the problem was not detectable without a working prototype, and the solution required a new screen rather than a modification to an existing one.



Figure 14. The transition screen introduced following the think-aloud evaluation of Flow 1b. Before back-up configuration begins, the screen addresses the care recipient directly and makes the control logic explicit: the back-up contact only becomes active when the patient decides, and must accept before gaining access (DP-09, DP-10).

The system limit identified as DP-25 is represented explicitly within this sub-flow. When there is no contingency network to configure, the flow surfaces a direct pathway to human clinical support rather than leaving the patient at a structural dead end. CG05's formulation "We have nobody. That is the problem, I have nothing to configure"[43] is the boundary condition this pathway acknowledges without pretending to solve.

[43] Original: *"Nosotros no tenemos a nadie. Ese es el problema, que no tengo nada que configurar."*(CG05, Workshop 4, Spain)

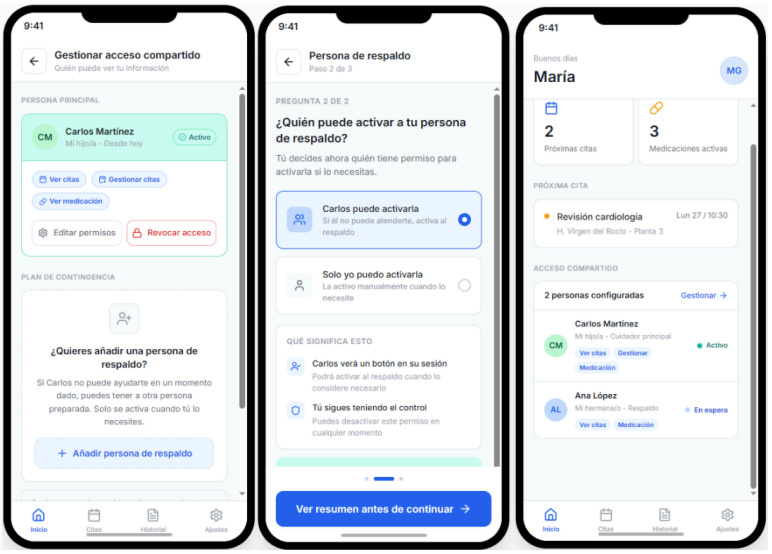


Figure 15. Flow 1b overview: the delegation management screen showing the primary caregiver's active status and the contingency configuration option (left), and the resulting dashboard with two configured caregivers in differentiated roles and access states (right) (DO4, DP-15).

Flow 2: modifying and revoking access. The expert interview with CG01 established the single strongest institutional argument of the entire project: revocation of delegated access is technically trivial. Removing the database correlation between a patient account and a delegated user would cause that access to disappear immediately. It was never designed as a user-facing function not because it could not be, but because it was not imagined as necessary. Flow 2 is therefore not speculative; it is the implementation of something the system could already do.

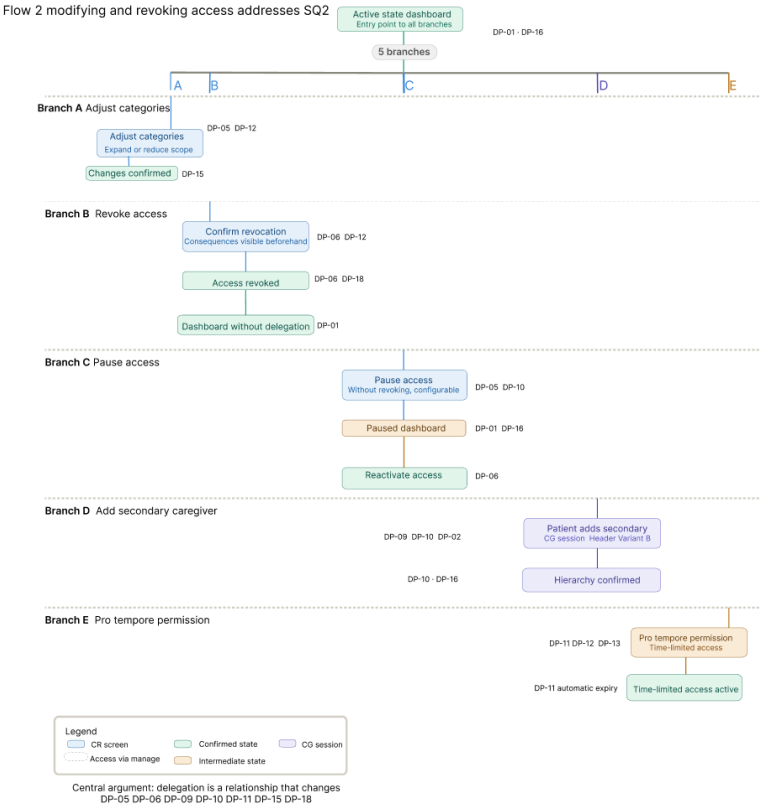


Figure 16. Interaction flow map for Flow 2 (modifying and revoking access). The flow covers five branches from a single entry point: adjusting categories, revoking access, pausing access, adding a secondary caregiver, and pro tempore permission.

The flow covers three distinct operations: modifying the scope of an existing delegation, pausing access temporarily, and revoking it entirely. The distinction between pausing and revoking was among the most clearly confirmed decisions in the think-aloud evaluation. CG011 articulated it without prompting:

> "These are two different things [...] here you are pausing, denying or granting access. You are not configuring."[44]

The revocation screen uses a red visual signal with a confirmation step. CG011 described the weight of the colour as appropriate, noting that it made the seriousness of the action legible without making the action inaccessible.[45]

Cascaded delegation is available only within rules set by the patient. A primary informal caregiver cannot extend their own access unilaterally (DP-10). Flow 2 makes this constraint visible: the secondary caregiver appears in the patient's delegation panel, not only in the primary caregiver's.

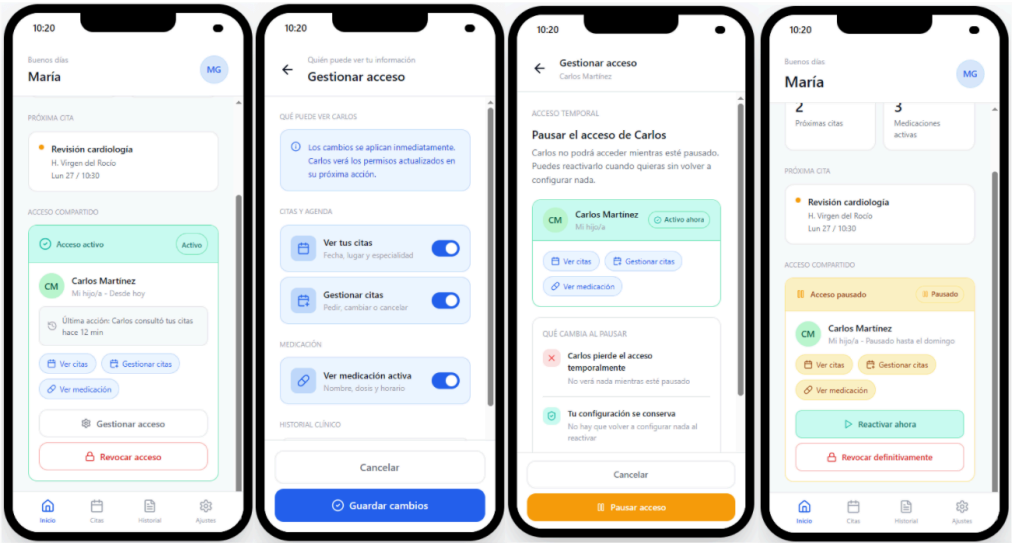


Figure 15. Selected screens from Flow 2 (modifying and revoking access). From left to right: the patient dashboard showing revocation as a persistent and accessible control (DP-15); the permission modification screen with immediate-effect toggles (DP-05); the pause screen distinguishing temporary suspension from permanent revocation while preserving the existing configuration (DP-09); and the paused-state dashboard showing the distinction between reactivating and revoking definitively.

Flow 3: acting and contributing. Flow 3 is structured around two sub-flows. Sub-flow 3a covers the caregiver's everyday delegated actions within the system, operating under their own authenticated identity with a persistent visual marker indicating they are acting on behalf of the care recipient. Sub-flow 3b covers clinical contribution through the observation log, providing the channel that every participant in the contextual inquiry phase had constructed by hand and no system in the sample provided. . When CG01 described the current system: "If I request on my mother's behalf, the system thinks my mother is active", he named exactly what the distinct caregiver session in sub-flow 3a is designed to correct.

[44] Original: "Sono due cose diverse. Anche se hai configurato ciò che la persona desidera, qui stai mettendo in pausa, negando o concedendo l'accesso. Non stai configurando. Sono due flussi diversi." (CG 011, Think-aloud Session 2, Italia)
[45] Original: "Mi fa capire che si trattava di qualcosa di importante." (CG 011, Think-aloud Session 2, Italia)

The clinical team receives a corresponding marker identifying whether an action was performed by the patient or by a delegated informal caregiver (DP-02) .

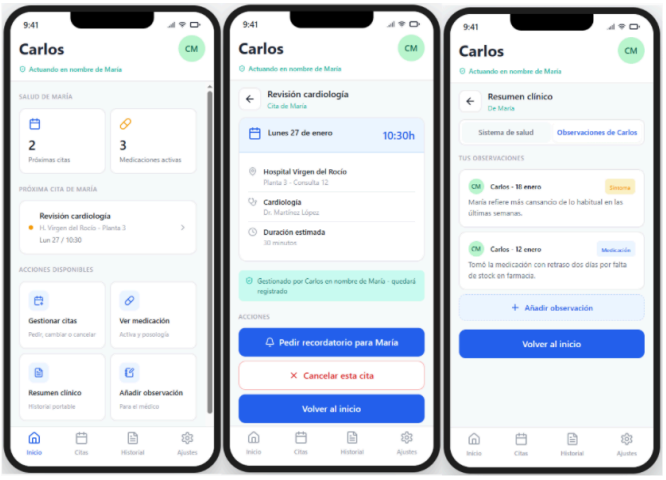


Figure 16. Selected screens from Flow 3 (acting and contributing). From left to right: the informal caregiver's distinct session dashboard, showing persistent attribution while acting on behalf of the care recipient (DP-01, DP-02); the appointment management screen showing caregiver attribution on every action (DP-02); and the two-author clinical summary, with system data and caregiver observations as co-equal inputs (DP-21).

CG04's formulation was the most direct: "I will keep the notebook regardless of what the app offers. What I want is for what I write to reach the doctor." The observation log accepts structured informal caregiver input, attributed to its author and visible to the clinical team before or during an appointment. Following DP-19 and DP-20, the informal caregiver is positioned throughout as an observer who records, not a clinician who interprets. The portable clinical summary has two co-equal authors: patient-generated data and informal caregiver observations, neither overwriting the other.

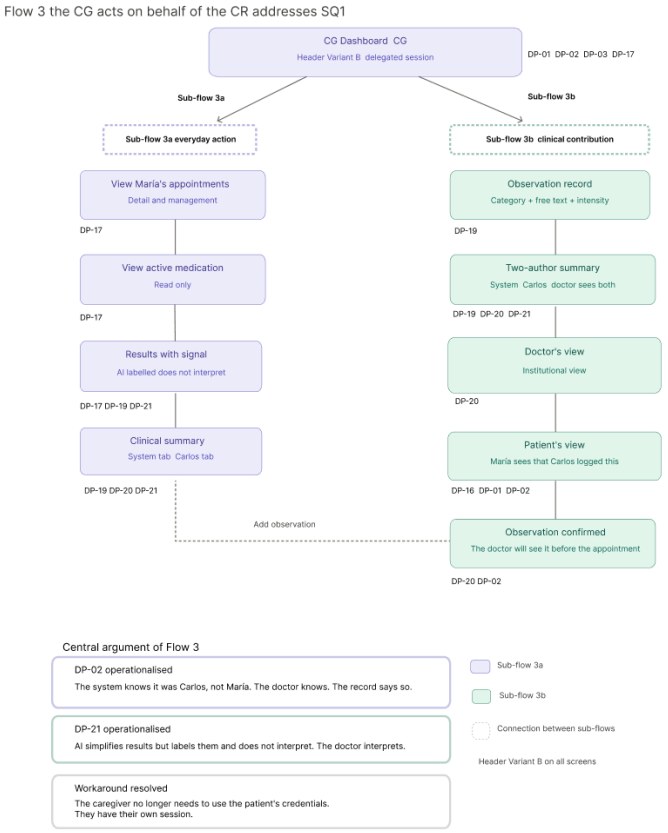


Figure 17. Interaction flow map for Flow 3 (acting and contributing). Sub-flow 3a covers everyday delegated actions; sub-flow 3b covers clinical contribution through the observation log. The two sub-flows connect at the clinical summary screen.

5.3 What the Evaluation Revealed

The think-aloud sessions placed the prototype in front of five participants across three dyads, covering all three flows in full. What they confirmed at the level of concept and what they contested at the level of implementation are equally part of the contribution.

The clearest confirmation concerns DP-15, which the workshops had established as a safety mechanism and which the evaluation reframed as a trust mechanism. CR07's response to the confirmation screen in Flow 1a[46] led directly to a formulation the workshops had not produced: “If I know I can take it back, I can give more.”[47] Revocability here is not a safety valve. It is what makes a broader scope of delegation feel possible in the first place. The design does not generate trust by restricting what the informal caregiver can see; it generates trust by ensuring the patient can change their mind.

The most important single finding from the evaluation concerns the observation log in Flow 3. CG011, faced with a field asking him to categorise an observed symptom as mild, moderate, or concerning, articulated the epistemic limit of the informal caregiver's role with clinical precision:

[46] Original:”Tu información sigue siendo tuya. Puedes revocar el acceso en cualquier momento desde tu panel. [...] Es clarísimo.”(CR07, Think-aloud Session 1, France)
[47]Original:”Si sé que puedo quitárselo, puedo dar más.” (CR07, Workshop 1, France)

> “I don't know whether it's something serious or moderate or concerning. Perhaps I can express the pain or discomfort that María is experiencing. I can ask her. But I don't feel I can assess it.”[48]

This finding could not have been obtained from a workshop. It required being present in front of a real form, in the role of informal caregiver, facing a decision with perceived clinical stakes. It proves that DP-07 and DP-13 are correctly formulated as principles but were violated in this specific implementation. The severity classification was removed. The informal caregiver describes; the clinical team interprets.

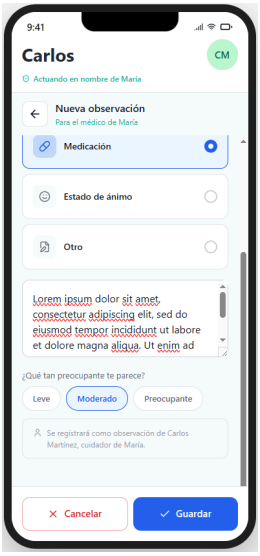


Figure 17. The observation log screen as presented during the think-aloud evaluation, showing the severity classification (mild / moderate / concerning, originally in Spanish: leve / moderado / preocupante) that was subsequently removed. CG11 refused to complete this field on the grounds that assessing clinical severity fell outside the informal caregiver's role. The final prototype removes the classification entirely, positioning the informal caregiver as an observer who describes rather than interprets (DP-19, DP-20).

The third finding is a tension the design must name rather than resolve. The disabled-default permission toggles produced opposite responses: CG08 found them a useful starting point that reduced cognitive load; CG04 experienced them as the system decided on her behalf. “It is as if the app were deciding for me.”[49] Both responses are correct, and they reflect care relationships with different histories and different dynamics. The revised prototype offers three named starting configurations, none pre-selected, with the rationale for each visible before the patient chooses. This does not dissolve the tension; it makes the decision conscious rather than implicit.

5.4 What the Design Cannot See

The most consistent finding across the complete dyads in this study is one that does not appear in any flow diagram or design principle: the research session was the first occasion on which most participants had discussed the delegation of digital health access directly. Workshop 1 surfaced this with precision. CR07 and CG08 had shared a caregiving

[48]Original:”Non so se è una cosa grave o moderata o preoccupante. Forse posso esprimere il dolore o il malessere che María sta vivendo. Posso chiederle. Ma non mi sento di valutare.” (CG 011, Think-aloud Session 2, Italy)

[49]Original:”Yo lo pondría todas en gris y ya el que quiera que lo vaya cambiando al azul. Porque así me parece como que me siento un poco obligada a que siga así. Pero es como si la aplicación estuviese decidiendo por mí.” (CG04, Think-aloud Session 3, Spain)

relationship for fifteen years. When the co-design session introduced the question of formal access, the exchange that followed was not a design discussion; it was a conversation that had not happened before:

> "I didn't tell you." / "You didn't want to talk about it. You were unwell. And when you are well you don't want to talk about when you are unwell."[50]

This pattern was not specific to that dyad.[51] Across CG07 and CR01, CG01 and CR02, CR05 and CG011, CG03, and CG06, the arrangements of informal care had been operating for months or years without the boundaries, conditions, or scope of that care ever having been explicitly named. The system had never created the conditions for that conversation to occur. Neither had the care relationship itself.

The design this project proposes responds to this directly. The onboarding flow is not a configuration form; it is a facilitated conversation. CG08 articulated this requirement in the think-aloud evaluation: the system should create the conditions for the conversation, with questions, with time, without rushing (DP-12). But naming this as a design response requires also naming what precedes it: the delegation conversation that research made possible is evidence not only of a design gap but of an affective one. People do not discuss the boundaries of care when doing so means confronting the reality that care is needed. The design cannot substitute for that confrontation. It can, at most, make the conversation less avoidable.

The case of CR08 makes this dimension of the problem visible in a form that the design both addresses and cannot fully resolve. CR08's husband navigates the Italian health system not through formal recognition of their marriage but through the system's unquestioned assumption that he is CR08's son. CR08 describes the result with precision:

> "Everyone, when we are out and about, everyone takes it for granted that he is my son."[52]

And:

> "That is how I feel at peace, having him close."[53]

That tranquillity is functional ignorance:[54] access permitted not because the relationship has been evaluated but because it has not been questioned. CR08 is a retired doctor with full institutional competence and decades of experience navigating the same system from the other side.[55] The invisibility of his informal caregiving relationship is not a consequence of digital exclusion. It is a structural condition produced by a system that was designed for a model of family that does not include his.

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[50] Original:*"No me lo dijiste."/"No querías hablar de ello. Estabas mal. Y cuando estás bien no quieres hablar de cuando estás mal."* (CR07 / CG08, Workshop 1, France)

[51] The pattern was documented across the sample. CG07 and CR01 had not discussed what would happen if CG07 became unavailable; CR01 confirmed this directly when asked. CG01 and CR02 operated on an implicit assumption of trust that had never been named. CR05 and CG011, CG03, and CG06 each described caregiving arrangements that had never been explicitly negotiated. In every case, the research session was the first occasion on which the question had been raised.

[52] Original:"Tutti, quando stiamo in giro, tutti danno per scontato che sia mio figlio. Lo danno per scontato." (CR08, Phase 1, Italy)

[53] Original:*"Io così mi sento tranquillo, avendo lui vicino."* (CR08, Phase 1, Italy)

[54] The concept of functional ignorance is introduced here to describe a condition distinct from formal recognition: access is permitted not because the system has evaluated the relationship but because it has not questioned it. This differs from the contextual integrity violations Nissenbaum (2004) describes, in which inappropriate information flows are actively produced; here the flow is permitted by default, and the risk is latent rather than immediate. In the case of CR08, the risk takes a specific form: the access that sustains him rests on a misrecognition that he and his husband must actively maintain, and that dissolves in any context where the system asks rather than assumes.

[55] CR08 is a retired doctor. He manages his own appointments, knows the cardiologist's receptionist by name, and speaks directly with specialists. His digital health competence is high. The invisibility of his caregiving relationship is therefore not a function of digital exclusion or cognitive limitation. It is a structural condition produced by a system that was not designed to see his marriage.

The design proposed in this project addresses this through two mechanisms. The relational identification field accepts any designation the patient chooses, without requiring institutional recognition of the relationship. The delegation criterion is the patient's active consent, not the system's prior recognition of the bond through which care is given. This operationalises Article 4(2a) of Regulation (EU) 2025/327, which grants citizens the right to delegate health data access to other natural persons without specifying the nature of the relationship. What the design cannot guarantee is how that formalisation interacts with institutional contexts where staff continue to apply informal assumptions. The formal delegation record this design produces gives CR08's husband a verifiable right that does not depend on what health professionals happen to assume. It does not guarantee that the assumption will stop being made.

The think-aloud sessions produced a further set of evidence that the design learned from precisely because it could not have been anticipated. The perspective-shift failure in Flow 1b, where CG08 could not identify whose view she was operating from ("I was seeing this screen as if I were María (CR)") required a new screen rather than a modification to an existing one. The severity classification in the observation log, which CG011 refused to complete on the grounds that he could not make a clinical judgment ("I don't feel I can assess it"),[56] required removing a field entirely. CG04's rejection of the institutional label *"informal care giver"* as the name for her role required the relational field to remain open rather than closed. In each case the design was not confirmed; it was corrected. That is not a failure of the design argument. In RtD, the correction is the argument: the prototype produced knowledge that workshops alone could not generate, and that knowledge changed the design.

What the design cannot correct is the condition of the body that uses it. CG04 entered the think-aloud evaluation with a diagnosis of caregiver burnout syndrome.[57] She is also the participant with the most clinically sophisticated informal documentation system in the sample, the participant who articulated most clearly what the observation log must do, and the participant who designed the traffic-light urgency system proposed in Workshop 3. These are not separate facts about the same person. They are the same fact: the body that has absorbed the most load is also the body that has generated the most design knowledge, because necessity is not a metaphor here. The design principles this project produces were not generated by users evaluating a neutral system. They were generated by people navigating systems that were not designed for them, under conditions those systems did not account for, in bodies that the interaction model did not imagine.[58]

These are not limitations of the design. They are the findings that define its scope. The design can formalise a caregiving relationship that the system never recognised, make the informal caregiver visible to a system that was registering their actions as someone else's, give the patient tools to define the terms of care on their own and change them as those terms change, and create the conditions for a conversation that fifteen years of caregiving had not produced. What it cannot do is reach CG04 before she developed burnout syndrome, or give CR08's husband a right that staff will automatically honour, or create a contingency network for CG05 where none exists, or make the delegation conversation happen before the crisis that

[56] Original:"Esta pantalla yo la veía como si fuese yo María. Y estaba añadiendo otra persona de respaldo." (CG08, Think-aloud Session 1, Spain)
[57] Original:"*Yo tengo el síndrome del cuidador quemado.*" (CG04, Phase 1, Spain)
[58] The pattern was documented across the sample. CG07 and CR01 had not discussed what would happen if CG07 became unavailable; CR01 confirmed this directly when asked. CG01 and CR02 operated on an implicit assumption of trust that had never been named. CR05 and CG011, CG03, and CG06 each described caregiving arrangements that had never been explicitly negotiated. In every case, the research session was the first occasion on which the question had been raised.

finally forces it. Knowing where the boundary falls, and why, is part of what this project has produced.

Both levels rest on a premise that neither of them states directly. The systems this project studied were designed for a user who was young, capable, digitally fluent, and acting alone. The care recipient whose body trembles and cannot complete a multi-step authentication was not that user. The informal caregiver managing someone else's health under sustained cognitive load, and absorbing the friction a system that was not designed for them generates, was not that user either. And the caregiving relationships that fall outside the institutional categories these systems recognise were not legible to them at all. Dourish's (2004) insistence that interaction is always situated in specific bodies under specific conditions is not an abstract theoretical commitment in this project but the methodological lens that made the structural invisibility of informal caregiving legible as a design problem in the first place.[59]

5.5 Discussion: Findings in Relation to Prior Work

The results presented in sections 4 and 5 speak directly to the literature reviewed in section 2: the HCI research on caregiving and proxy participation, the bioethical frameworks for health information access, and the existing design implementations of delegated access. This section identifies where the findings confirm those accounts, where they extend them, and where they produce knowledge the literature has not anticipated.

The most consistent finding across all three fieldwork contexts, that informal caregivers have constructed sophisticated parallel documentation systems and routinely access health portals using the patient's own credentials, confirms and extends what prior work has documented in analogous settings. Latulipe et al. (2022) found that people helping older adults with online banking regularly use the older adult's credentials as a matter of necessity rather than choice; the present study finds the same pattern in health contexts, where a credential-sharing arrangement that creates privacy and security risks in banking can also compromise the integrity of a clinical record and expose both parties to GDPR violations neither of them chose. What this project adds to that account is twofold. First, the institutional mechanism that makes the workaround invisible: because the system registers every action taken from a patient's device as the patient's own, it never generates the evidence of a gap that might prompt a design response. Second, the finding that informal caregivers do not simply want a digital replacement for their analogue systems: CG04 stated explicitly that she would keep her notebook regardless of what the application offered, because the notebook functions as a risk management strategy rather than a workaround. What she needed was a channel for what she already documented to reach the clinical team, a distinction the design principles produced here attempt to operationalise.

That channel does not exist in any of the systems studied, and the internal documentation of the Andalusian Health System makes clear why. The ClicSalud v2.11 manual specifies that clinical information within the system originates exclusively from internal sources and cannot be introduced from external ones, a design decision that structurally forecloses exactly the contribution that participants like CG04 were making by hand. Moll et al. (2024) argue that the absence of delegation mechanisms is a design and policy choice rather than a

[59] Dourish (2004) argues that interaction is always situated in specific bodies in specific contexts under specific conditions. The research itself was shaped by a body navigating institutional systems from a position of partial non-recognition, what Núñez-Pacheco and Frid (2023) describe as autobiographical design: bringing personal experience into the research process as a commitment that shapes how data is collected, analysed, and interpreted.

technical limitation; the SAS documentation confirms that claim at the level of architectural specification and adds a dimension that comparative policy analysis cannot reach: the choice was not made once but is reproduced across four decades of system development, across every version of every manual, without the informal caregiver ever appearing as a named user. Scerpella et al. (2025) identify a recognisable ceiling in their work on shared access to patient portals: a tutorial can teach people how to use a system that was not designed with them in mind, but it cannot introduce the relational logic of caregiving into a system that was not built to support it. The SAS documentation confirms why that ceiling exists: it is not a usability problem but an architectural one.

Where the system produces a gap it cannot see, the caregiving relationship produces a negotiation the system cannot support. The finding that delegation operates as a continuous and negotiated process rather than a permission state configured once confirms and extends the theoretical framework developed by Lazar et al. (2018), who demonstrate that care relationships are ongoing negotiations in which the boundaries of appropriate access and the trust on which delegation depends are continuously renegotiated as circumstances change. The present study provides empirical evidence of what that negotiation looks like in a digital health context specifically: not a single configuration moment but a trajectory that may begin with supported use and shift toward substituted action over months or years, following the spectrum described by Storer and Branham (2023). A related finding extends what Latulipe et al. (2018) document about patient ambivalence: participants in this study did not want to give informal caregivers access to everything or nothing, but to specific things under specific conditions, with CR05 introducing a distinction between recent and historical diagnoses that no existing system accommodates, and CR04 articulating the underlying principle directly: the system must make visible that the caregiver helps, without displacing the patient as the person whose health it is. What neither Lazar et al. nor Storer and Branham fully anticipate is the finding that runs through every complete dyad: in every case, the research session was the first occasion on which patient and caregiver had discussed the scope of delegated digital health access directly, across relationships of five, ten, and fifteen years. The system had never created the conditions for that conversation to occur, and neither had the caregiving relationship itself. The onboarding flow this project proposes is a direct response to that finding: not a configuration form but a facilitated conversation, designed to make the delegation discussion less avoidable.

What that negotiation requires from a system, and what existing systems do not provide, is the ability to calibrate, adjust, and withdraw access in ways that reflect the specific norms of the caregiving relationship. The interaction qualities that participants consistently identified as necessary for trust, specifically granularity, revocability, and transparency, map directly onto the requirements that Meslin and Schwartz (2014) derive from bioethical analysis of electronic health record design: ethical compliance in delegated access requires designing for the patient's ability to calibrate, adjust, and withdraw it. The present study confirms this empirically and adds one finding the bioethics literature does not anticipate: revocability functions not primarily as a safety mechanism but as a trust enabler. CR07's formulation during the think-aloud evaluation, that knowing he could take access back made him willing to give more, provides the empirical grounding for what Nissenbaum (2004) argues theoretically about contextual integrity: that information should flow in ways that match the norms of the specific relationship it moves through. The existing implementations that come closest to these requirements, Estonia's granular permission controls and the NHS National Proxy Service, each address part of the problem without reaching the relational specificity

that Nissenbaum's framework requires and that the SAS documentation reveals has never been attempted: the system offers binary access or no access, with no mechanism for the contextual calibration that both the bioethics literature and the fieldwork consistently identify as the condition under which delegation can be trusted.

That relational specificity matters most in the caregiving relationships that existing systems are least equipped to recognise. Fredriksen-Goldsen et al. (2015) and Muraco and Fredriksen-Goldsen (2011) document that informal care in LGBTQ+ communities is most often provided by partners and friends rather than biological relatives, and that the absence of formal institutional recognition of those relationships creates specific vulnerabilities in moments of crisis. The case of CR08 in this study confirms that pattern in a digital health context and adds a specific mechanism: his husband navigates the Italian health system not through formal recognition of their marriage but through an institutional assumption that has never been questioned and that dissolves the moment the system stops assuming and starts asking. Weston's (1991) concept of chosen family describes exactly the caregiving structure that eHealth systems were not designed to accommodate. The delegation criterion this project proposes, the patient's active consent rather than the system's prior recognition of the relationship, responds to that structural gap and aligns with Article 4(2a) of Regulation (EU) 2025/327, which grants citizens the right to delegate health data access to other natural persons without specifying the nature of the relationship. That the interaction layer required to operationalise this right does not yet exist in any of the systems studied is precisely the gap this project has attempted to fill.

6. CONCLUSION

This project set out to understand how interaction with digital health systems can be designed to support the gradual and negotiated delegation of access between patients and their informal caregivers, within eHealth infrastructures where formal proxy mechanisms are absent or inadequate. Conducted through Research through Design across Spain, Italy, and France with twenty-three participants, the study produced findings that address the research questions at three levels: informal caregivers have constructed sophisticated parallel documentation systems in the absence of any institutional equivalent, answering SQ1; delegation operates as a continuous and negotiated process rather than a permission state configured once, answering SQ2; and the interaction qualities that generate trust, specifically granularity, revocability, and transparency, can be operationalised in concrete interaction flows, which constitutes the answer to SQ3 and the transferable claim this project makes.

The analysis of the three systems studied confirms that this absence is not a technical limitation but a design decision. Across Spain, Italy, and France, the informal caregiver was never imagined as a user. The Andalusian Health System documents this with particular precision: the workaround that informal caregivers use resolves the operational need without leaving a trace, making the system structurally blind to the GDPR violations it forces its users to commit and to the gap it produces.

These findings support two levels of contribution to interaction design for health information systems. The first is a vocabulary for what was previously unnamed. The informal caregiver who shares a patient's credentials is not committing a privacy violation out of negligence but responding rationally to a system that offers no legitimate alternative, and the system that registers every action taken from a patient's device as the patient's own is not malfunctioning

but functioning exactly as designed, for a user it imagined and against a user it did not. Naming this as unauthorised necessity rather than user error, and the informal caregiver's body as a design constraint rather than a usability problem, is the vocabulary this project contributes, one that did not exist in the systems studied before this work. The second level is the demonstration that the problem is designable: the three interaction flows, the twenty-five design principles, and the medium-fidelity prototype, corrected in three specific ways by the people it was designed for, advance the argument that delegation of digital health access can be supported when a system treats it as a relationship to be maintained over time rather than as a permission state to be configured once.

These two levels of contribution rest on a finding that runs through every complete dyad (every caregiving pair in which both patient and caregiver participated) in the study: in every case, the research session was the first occasion on which patient and caregiver had discussed the scope of delegated digital health access directly, across relationships of five, ten, and fifteen years. This is both a finding about the depth of the gap and evidence that the onboarding flow this project proposes can create the conditions for a conversation that years of caregiving had not produced.

Situating these contributions within their limits requires the same precision. The limits are of two kinds: those that fall within the scope of interaction design but were not fully resolved by this study, and those that fall outside it entirely, where interface design reaches the boundary of what it can change without institutional or policy intervention. The design principles and prototype address the interaction layer of delegation and demonstrate that granular, revocable, and relationally sensitive delegation is both technically feasible and responsive to correction by the people who would use it. What they do not resolve is the policy threshold that currently excludes the entire sample from formal delegation rights, the structural absence of contingency networks for caregivers without support, or the institutional non-recognition of caregiving relationships that fall outside legal kinship categories. These are the conditions that generated the most significant design constraints in this project, and they point directly toward the institutional and policy changes that Regulation (EU) 2025/327 has already mandated and that most member states have not yet designed for.

What this project contributes is therefore specific and transferable: twenty-five design principles, fifteen confirmed across participant profiles, delegation types, and national contexts; a prototype corrected by the people it was designed for; a confirmed institutional account of why revocation of delegated access was never designed as a user-facing function; and a vocabulary for a structural condition that the systems responsible for it cannot currently see. Those contributions, and the limits they carry, point toward three directions for future work: evaluation of the principles within systems actively implementing delegation under Regulation (EU) 2025/327, where testing against real implementations would establish which travel beyond the three contexts studied here; research into the institutional changes required to reach caregivers without contingency networks and those whose relationships fall outside legal kinship categories; and further investigation of the dyad as a unit of design, for which this project offers evidence but does not exhaust the question.

This project began with a structural problem that the fieldwork made immediate: the forty to fifty million informal caregivers across Europe who manage digital health systems on behalf of someone who cannot do so alone have no formal mechanism to make that role legitimate. The findings presented here offer a more precise account of why that gap persists: not

because the technical infrastructure is absent, but because the interaction design layer that would translate the needs of real caregiving relationships into system behaviour has not yet been built. What this project has attempted to demonstrate is that building it is possible, that the knowledge required can be generated with the people who need it, and that it can be made transferable beyond the contexts in which it was produced.

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To everyone who provides care for someone else: this project was made with you in mind, and I hope it is useful to whoever comes after.

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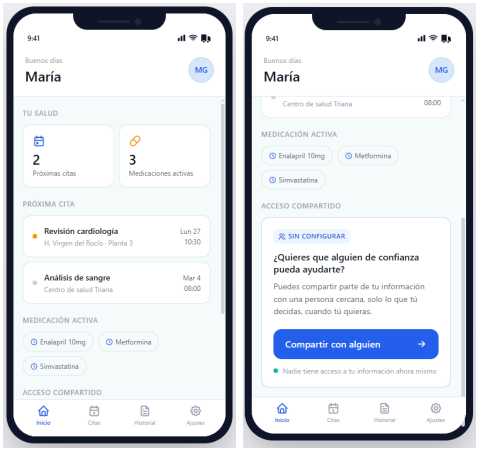
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Appendix A: Prototype Screens

Flow 1: Configuring Delegation Flow

Flow 1a: Setting up shared access for the first time (configuración inicial del acceso compartido)

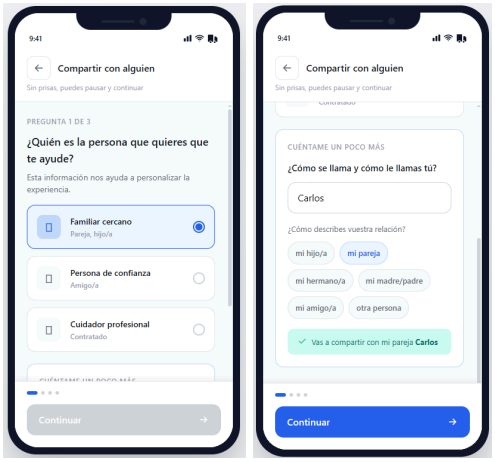
Screen 1



Screen 1 (left): Patient home screen (inicio). Displays a summary of upcoming appointments (próximas citas), active medications (medicación activa), and a bottom navigation bar. No shared access configured yet. Flow 1a.

Screen 1 (right): Scrolled state of the home screen. The "Shared access" (acceso compartido) section becomes visible with a call-to-action prompting the patient to share their information with a trusted person. A "Share with someone" (compartir con alguien) button initiates the delegation flow. Flow 1a.

Screen 2.

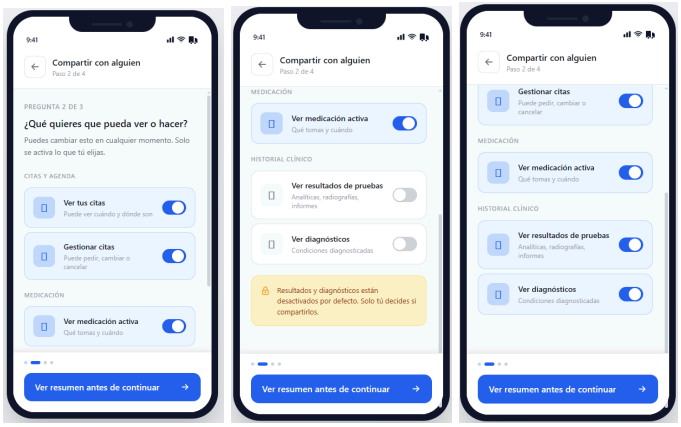


Screen 2 (left): Onboarding question 1 of 3. The patient selects the type of person they want to share access with. Options are: close family member (familiar cercano), trusted person (persona de confianza), and professional caregiver (cuidador profesional). "Close family member" is selected. The continue button remains inactive. Flow 1a.

Screen 2 (right): Expanded follow-up section "Tell me a little more" (Cuéntame un poco más). The patient enters the caregiver's name ("Carlos") and selects the relationship type

("my partner" / mi pareja). A confirmation summary appears at the bottom. The continue button becomes active. Flow 1a.

Screen 3

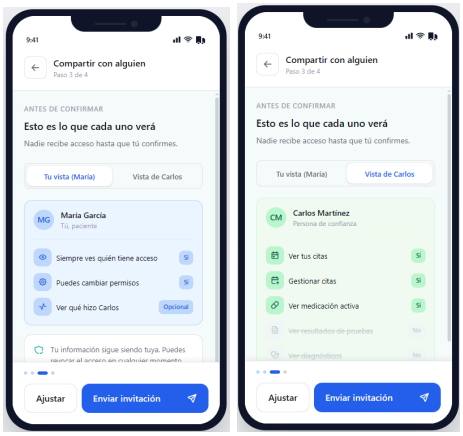


Screen 3 (left): Onboarding question 2 of 3. The patient selects what the caregiver can see or do (qué quieres que pueda ver o hacer). Permissions are grouped by category. Under appointments (citas y agenda): view appointments (ver tus citas) and manage appointments (gestionar citas) are enabled by default. Flow 1a.

Screen 3 (center): Scrolled state of Screen 3. Under clinical history (historial clínico): view test results (ver resultados de pruebas) and view diagnoses (ver diagnósticos) are disabled by default. A warning notice highlights that these are off by default and only the patient can decide to share them. Flow 1a.

Screen 3 (right): Scrolled state showing all permissions enabled, including view test results and view diagnoses under clinical history. All toggles are active. Flow 1a.

Screen 4

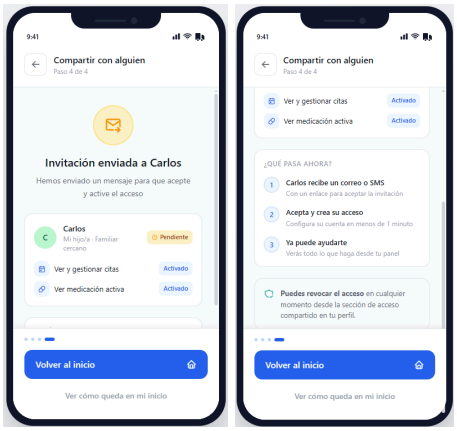


Screen 4 (left): Pre-confirmation summary, step 3 of 4. The patient reviews what each person will see (esto es lo que cada uno verá). The CR's view (tu vista / María) is active, showing retained controls: always seeing who has access, being able to change permissions, and optionally viewing CG's (Carlos) activity log. Flow 1a.

Screen 4 (right): Same pre-confirmation summary with the informal caregiver's view (vista de Carlos) selected. Shows the permissions granted to CG (Carlos): view appointments (ver

tus citas), manage appointments (gestionar citas), and view active medication (ver medicación activa). Test results and diagnoses are marked as not shared. Flow 1a.

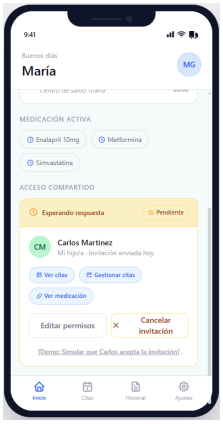
Screen 5



Screen 5 (left): Confirmation screen, step 4 of 4. The invitation has been sent to CG (Carlos) (invitación enviada a Carlos). Shows CG's (Carlos) access status as pending (pendiente) with enabled permissions listed. The patient can return to the home screen or preview how it looks there. Flow 1a.

Screen 5 (right): Scrolled state of Screen 5. A "What happens now?" (¿qué pasa ahora?) section explains the next steps: CG (Carlos) receives an email or SMS, accepts and creates his access, and can then start helping. A reminder informs the patient that access can be revoked at any time. Flow 1a.

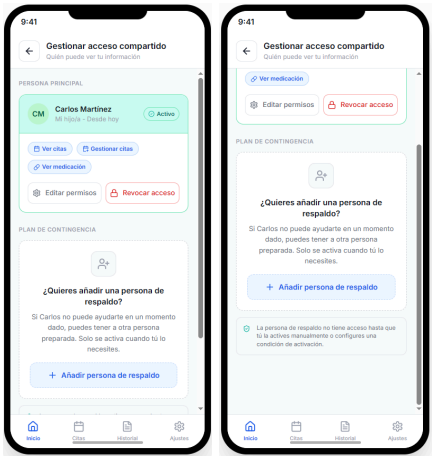
Screen 6



Screen 6: Home screen after invitation sent. The shared access (acceso compartido) section shows CG's (Carloss) invitation as pending (esperando respuesta / pendiente), with granted permissions listed as tags. The patient can edit permissions or cancel the invitation. A demo trigger is visible for prototype testing purposes. Flow 1a.

Flow 1.b:Adding a backup person to the delegation setup (añadir persona de respaldo)

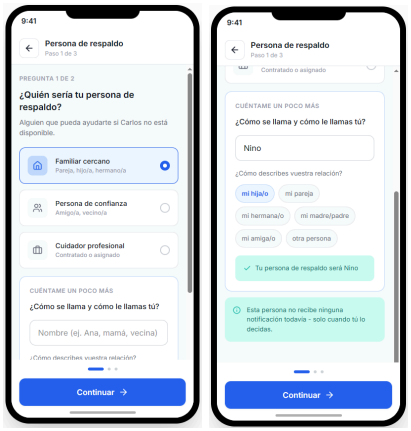
Screen 1.b.1



Screen 1.b.1 (left): Shared access management screen (gestionar acceso compartido). Shows Carlos Martínez (CG) as the primary person (persona principal) with active status, granted permissions listed as tags, and options to edit permissions or revoke access. A contingency plan (plan de contingencia) section prompts the patient to add a backup person (persona de respaldo). Flow 1b.

Screen 1.b.1 (right): Scrolled state of Screen 1 b 1. The contingency plan section is fully visible, explaining that the backup person only gains access when the patient manually activates it or sets an activation condition. Flow 1b.

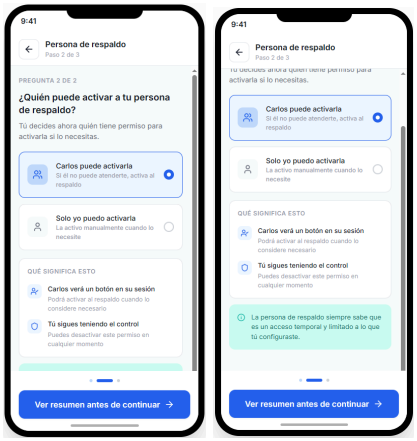
Screen 1.b.2



Screen 1.b.2 (left): Backup person setup, step 1 of 3. The patient selects the type of backup person (¿quién sería tu persona de respaldo?). "Close family member" (familiar cercano) is selected. The "Tell me a little more" (cuéntame un poco más) section is visible but the name field is still empty. Flow 1b.

Screen 1.b.2 (right): Scrolled state of Screen 2. The patient has entered the backup person's name ("Nino") and selected the relationship "my daughter/son" (mi hija/o). A confirmation summary appears, along with a notice that the backup person will not receive any notification until the patient decides. Flow 1b.

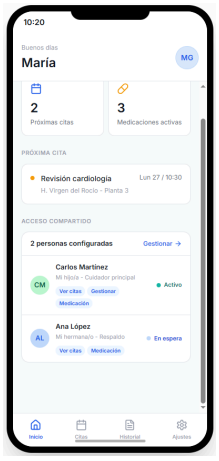
Screen 1.b.3



Screen 1.b.3 (left): Backup person setup, step 2 of 3. The patient defines who can activate the backup person (¿quién puede activar a tu persona de respaldo?). "Carlos (CG) can activate it" (Carlos puede activarla) is selected. A "What this means" (qué significa esto) section explains that CG (Carlos) will see a button in his session and that the patient retains control at all times. Flow 1b.

Screen 1.b.3 (right): Scrolled state of Screen 3. A notice confirms that the backup person will always know their access is temporary and limited to what the patient configured. Flow 1b.

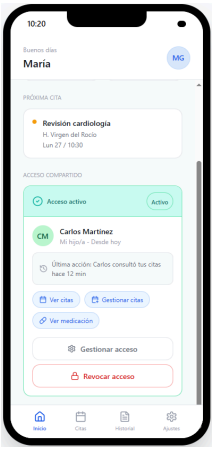
Screen 1.b.4



Screen 1.b.4: Home screen after backup person configured. The shared access (acceso compartido) section shows two people configured: Carlos Martínez (CG)as the primary informal caregiver (cuidador principal) with active status, and Ana López as the backup person (respaldo) with pending status (en espera). Granted permissions are listed as tags for each person. Flow 1b.

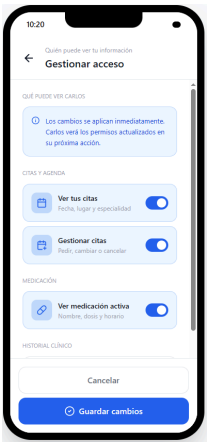
Flow 2: Modifying and Revoking Access

Screen 1.2



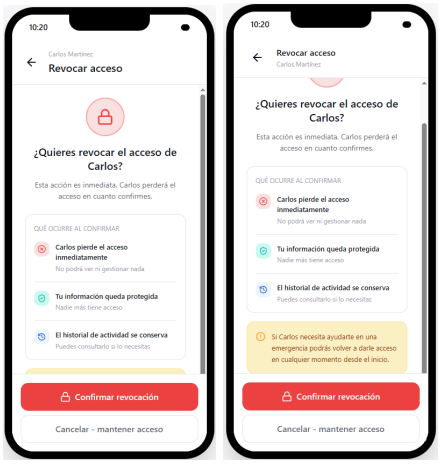
Screen 1.2: Home screen with active shared access. The shared access (acceso compartido) section shows CG's (Carlos) access as active, including the last action performed and granted permissions as tags. Options to manage or revoke access are visible. Flow 2.

Screen 2.2



Screen 2.2: Permission management screen (gestionar acceso). Shows current permissions grouped by category with toggles. A notice informs that changes apply immediately. Options to cancel or save changes at the bottom. Flow 2.

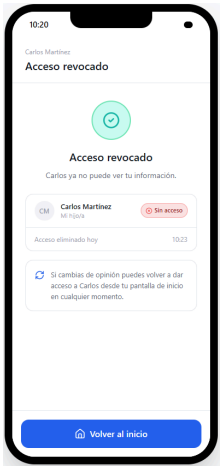
Screen 3.2



Screen 3.2 (left) : Access revocation screen (revocar acceso). The patient is asked to confirm revocation for CG (Carlos). A "What happens on confirm" (qué ocurre al confirmar) section lists the consequences: immediate loss of access, information remains protected, and activity log is preserved. Flow 2.

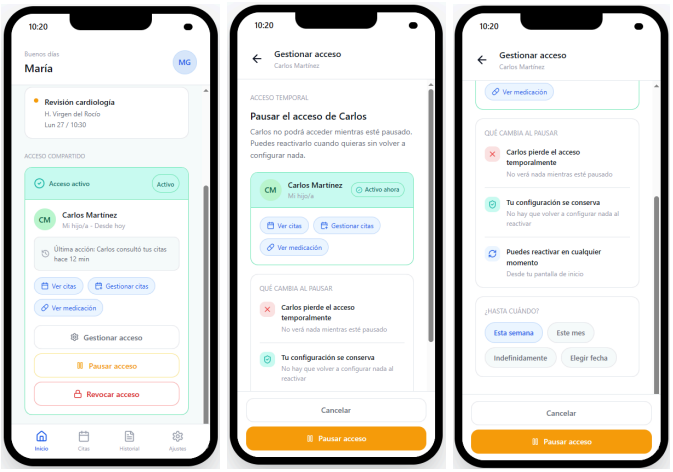
Screen 2.4 (right): Scrolled state of Screen 3. A warning notice reminds the patient that access can be reinstated at any time from the home screen in case of emergency. Confirm revocation and cancel buttons are visible. Flow 2.

Screen 4.2



Screen 4.2 : Confirmation screen after revocation. CG's (Carlos) status is shown as no access (sin acceso) with the timestamp of removal. A notice reminds the patient that access can be restored at any time from the home screen. Flow 2.

Flow 2.b: Pausing and revoking access
Screen 1.2b



Screen 1.2b (left): Home screen with active shared access. The shared access (acceso compartido) section shows CG's (Carlos)access as active, including the last action performed and granted permissions as tags. Options to manage (gestionar), pause (pausar) or revoke (revocar) access are visible. Flow 2b.

Screen 1.2b (center): Pause access screen (pausar el acceso de Carlos,CG). Shows current active permissions and explains what changes when paused: Carlos loses access temporarily and the configuration is preserved for reactivation. Flow 2b.

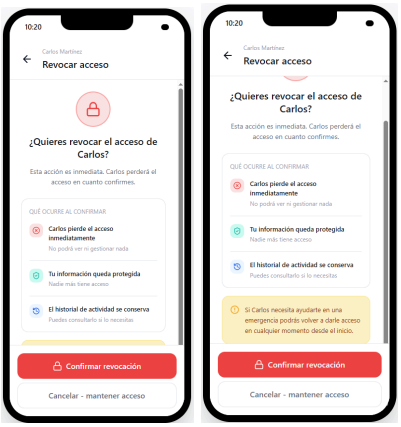
Screen 1.2b (right): Scrolled state of Screen 1a. A "Until when?" (¿hasta cuándo?) section allows the patient to select the pause duration: this week (esta semana), this month (este mes), indefinitely (indefinidamente), or a custom date (elegir fecha). Flow 2b.

Screen 2.2b



Screen 2.2b: Home screen after access paused. The shared access section shows CG's (Carlos) status as paused (acceso pausado) until Sunday. Options to reactivate immediately (reactivar ahora) or revoke permanently (revocar definitivamente) are visible. Flow 2b.

Screen 3.2b



Screen 3.2b (left): Access revocation screen (revocar acceso) reached from the paused state. Confirms consequences of revocation: immediate loss of access, information remains protected, and activity log is preserved. Flow 2b.

Screen 3.2b (right): Scrolled state of Screen 3. A warning notice reminds the patient that access can be reinstated at any time from the home screen in case of emergency. Flow 2b.

Screen 4.2b



Screen 4.2b: Confirmation screen after revocation. CG's (Carlos) status is shown as no access (sin acceso) with the timestamp of removal. A notice reminds the patient that access can be restored at any time. Flow 2b.

Flow 3: The caregiver acts on behalf of the patient (CG acts on behalf of CR)

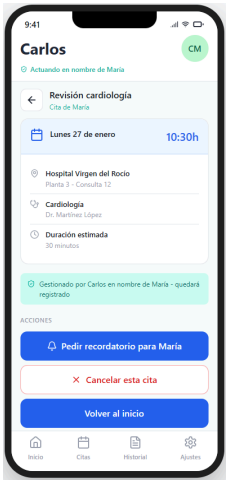
Sub-flow 3a: Everyday actions

Screen 1.3a



*Screen 1.3a:*Caregiver home screen. Carlos accesses María's health information acting on her behalf (actuando en nombre de María). Shows upcoming appointments and active medications summary, and available actions: manage appointments (gestionar citas), view medication (ver medicación), clinical summary (resumen clínico), and add observation (añadir observación). Flow 3a.

Screen 2.3a



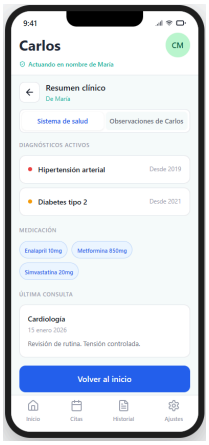
Screen 2.3a : Appointment detail screen. Carlos views the cardiology appointment details including date, time, location, and doctor. Actions available: request a reminder for María or cancel the appointment. A notice confirms the action will be logged. Flow 3a.

Screen 3.3a



Screen 3.3a: CR's active medication screen (medicación de María). Read-only view showing active medications with dosage and schedule. A warning clarifies that Carlos can view but not modify medication. Flow 3a.

Sub-flow 3b: Clinical contribution
Screen 1.3b



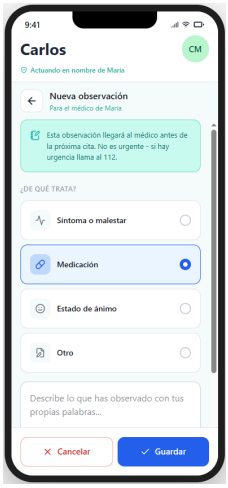
*Screen 1.3b:*Clinical summary (resumen clínico), health system tab (sistema de salud) active. Shows CR María's active diagnoses, current medications, and last consultation. Flow 3b.

Screen 2.3b



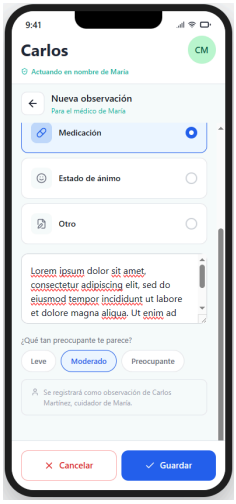
*Screen 2.3b:*Clinical summary, Carlos's observations tab (observaciones de Carlos) active. Lists previous observations added by Carlos, tagged by category. Option to add a new observation. Flow 3b.

Screen 3.3b



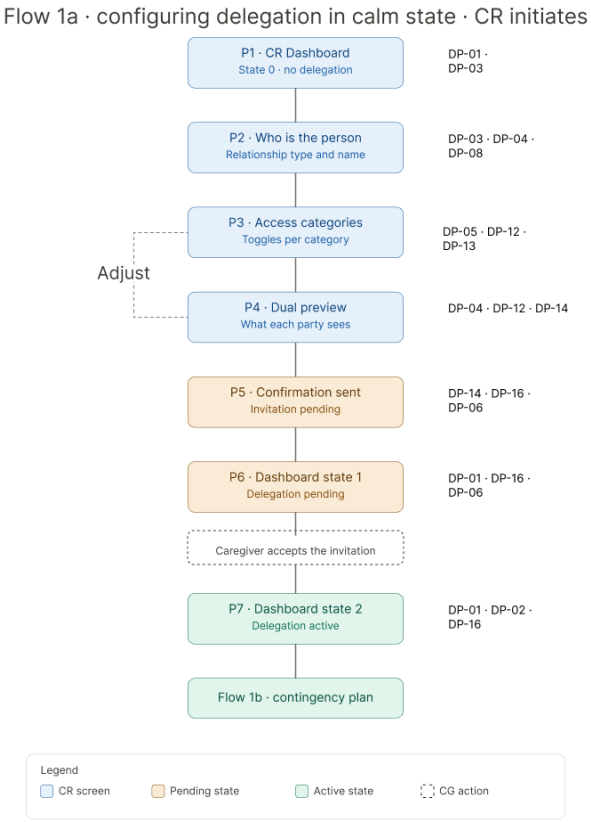
Screen 3.3b: New observation screen (nueva observación). CG (Carlos) selects the observation category and writes a free text description. A notice clarifies this is not urgent. Flow 3b.

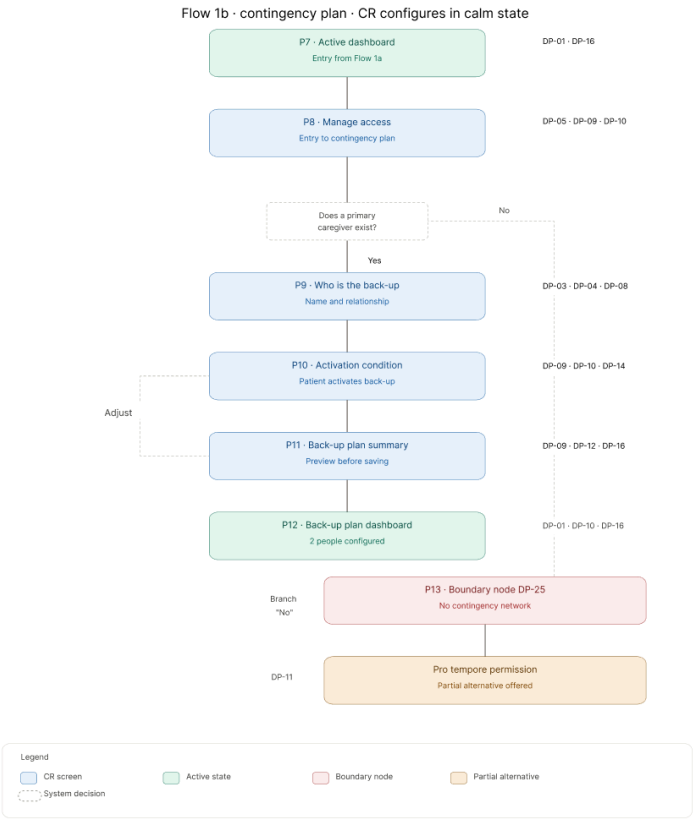
Screen 4.3b



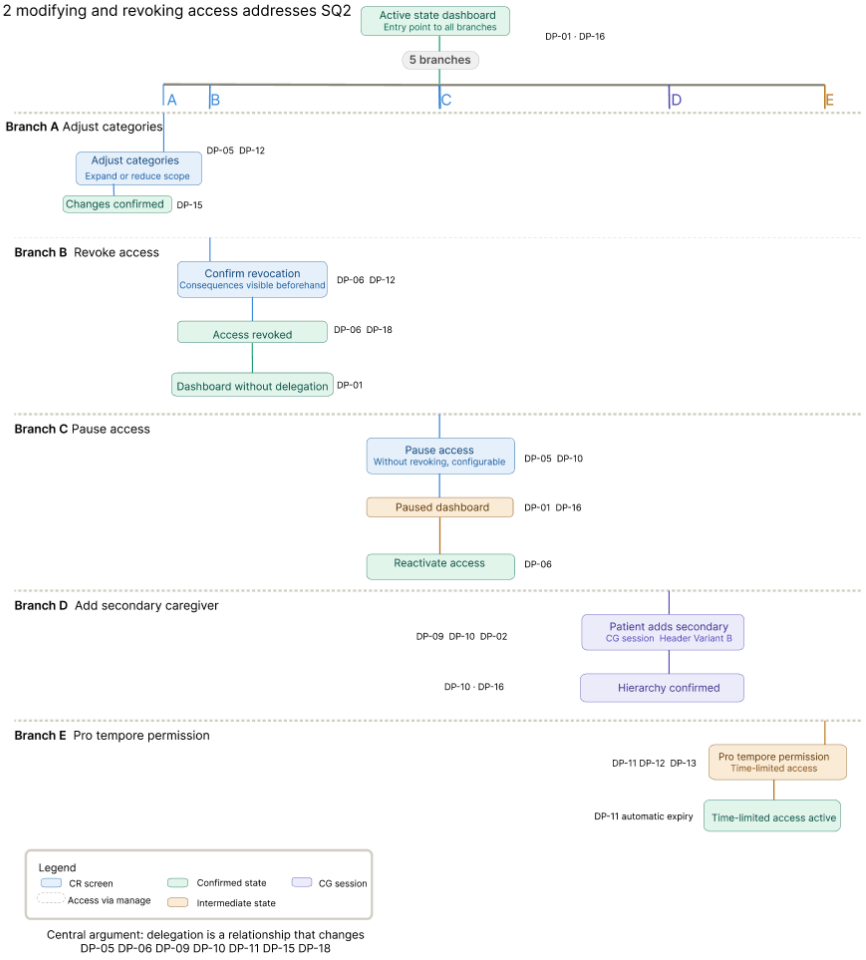
Screen 4.3b: Scrolled state of Screen 6. Carlos has filled in the observation text and selected severity level "moderate" (moderado). A notice confirms the observation will be registered under Carlos's name as María's caregiver. Flow 3b.

Appendix B: Interaction Flow Maps



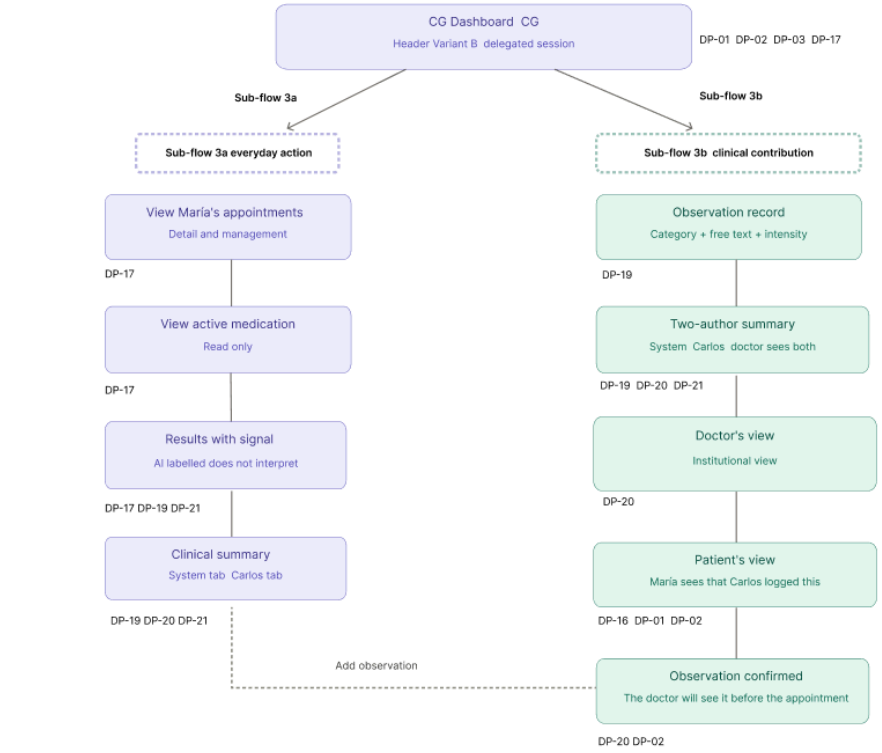


Flow 2 modifying and revoking access addresses SQ2



Central argument: delegation is a relationship that changes
DP-05 DP-06 DP-09 DP-10 DP-11 DP-15 DP-18

Flow 3 the CG acts on behalf of the CR addresses SQ1



Central argument of Flow 3

DP-02 operationalised

The system knows it was Carlos, not María. The doctor knows. The record says so.

DP-21 operationalised

AI simplifies results but labels them and does not interpret. The doctor interprets.

Workaround resolved

The caregiver no longer needs to use the patient's credentials.
They have their own session.

Sub-flow 3a

Sub-flow 3b

Connection between sub-flows

Header Variant B on all screens